

DRO Quick Reference Guide - CSP

Accessing DRO	Obtaining Technical Support
DRO is accessed via https://dro.routesmart.com	To obtain technical support, please call 1-800-ROUTING
Choose the Contract Service Provider login	
Enter your credentials for access to DRO	
For more information about any of the sections in this Reference Guide, see the DRO user guide.	

Fine Tune Route Plans

Use DRO tools to create Stop Overrides, Anchor Areas, and Breaks to add more detailed requirements for your routes.

Stop Overrides - The CSP can specify stop overrides in DRO to force specific stops to behave in a certain way. Features of a Stop Override include:

- **Route Assignment** - Stop must be assigned to a particular route or route type
 - Stop can be assigned to a Bulk Route (free to be on any Bulk route)
 - Stop can be assigned to an LP Route (free to be on any LP route)
 - Stop can be assigned to a specific Work Area (must be assigned to the vehicle servicing the work area)
 - No route assignment – Only valid if assigning the stop a First / Last order below
- **First / Last** - Stop must be done at the beginning or end of a route
 - Stop can be designated to be serviced first or last on the route
 - Multiple first/last designations on a route will be done in the most efficient order
- **Load Section** - Items for stop may be assigned to specific section of a vehicle
 - Can designate particular shelf or floor section, or no preference
 - No preference stops will be placed in the vehicle in sequence order

Anchor Areas - The CSP can specify “Anchor Areas” in DRO to encourage stops in a certain geographic area to behave in a certain way.

Breaks (optional) - Indicate the length of break, as well as earliest and latest start times

Use Fine Tuning

Edit existing route plans to make use of stop overrides and anchor areas.

- Global constraints:
 - Stop overrides
 - Neighborhood clusters
 - Add/remove vehicles
- Route-level constraints:
 - Wave #

- Start time
- Target hours
- Anchor areas

Generate Routes during planning window

During the planning window, you can create routes using the active route plan for the day for the current set of waypoints.

Swap stops to refine routes

To fine-tune your routes, you may find it necessary to move some points from one route to another.

View reports

Review multiple reports for your Actual and/or Informed Optimal solutions.

Export map (optional)

The Export feature allows you to save your map as a pdf or image file.

View/Review dashboard

Review Package Assignment Summary, Routing Status, and Unrouteable Stops for the CSA.



DRO CSP User Guide

Table of Contents

Dashboard	4
Map	6
Display Options	6
Map Tools	6
Stop Information Table Tools	7
Map Functions	8
Reports	14
Managing Communications	16
Adding Contacts to the Global Contact List	16
Editing a Contact	16
Deleting a Contact	17
Editing Alerts	17
Managing Your Fleet	19
Adding a Driver	19
Editing a Driver	20
Deleting a Driver	21
Adding a Vehicle	22
Customizing Section Availability	24
Editing a Vehicle	25
Copying a Vehicle	26
Deleting a Vehicle	27
Adding a Vehicle Cost Profile	27
Editing a Vehicle Cost Profile	28
Deleting a Vehicle Cost Profile	28
Adding a Custom Load Profile	29
Editing a Custom Load Profile	30
Deleting a Custom Load Profile	30
Managing Route Plans	31
Creating a Route Plan	31
Editing a Route Plan	32
Editing Relays in a Route Plan	37
Editing Start and End Points by Route	38
Copying a Route Plan	38
Deleting a Route Plan	39
Scheduling Your Route Plans	39
Managing Anchor Areas	40
Adding an Anchor Area	40
Renaming Anchor Areas	41
Reshaping an Anchor Area	41
Assigning Anchor Areas to Routes	43
Anchor Area Sequencing and Prioritizing	44
Showing an Anchor Area on the Map	48
Editing an Anchor Area	48
Deleting an Anchor Area	48

Using Dynamic Anchor Areas	49
Enabling Dynamic Anchor Areas	49
Creating Dynamic Anchor Areas	50
Sequencing Dynamic Anchor Areas	52
Managing Stop Overrides	55
Adding a Stop Override	56
Adding a Stop Override on the Map Page	58
Editing a Stop Override	59
Deleting a Stop Override	60
Exporting Stop Overrides to csv	60
Managing Street Settings	61
Delivery Side	61
Street Closures	63
Viewing Streets Only	65
Managing Defaults	67
Editing Bulk Stop Customization	67
Editing Large Package Customization	67
Editing Small Stop Customization	69
Editing Floor Loading Rules	69
Adding Driver Breaks	70
Editing Driver Breaks	71
Deleting a Driver Break	71
Enabling Relays	71
Viewing Historical Volume Information	72
Using the Sandbox Environment	73
Accessing the Sandbox Environment	73
Syncing from Production	74
Copying a Route Plan	74
Adding New Work Areas	74
Using Sort Plan View	76
Accessing Sort Plan View	76
Using the Manifest View	77
Accessing Manifest View	77
Exploring the Map in Manifest View	77
Exploring the Reports Page in Manifest View	78

Dashboard

On the Dashboard in DRO, you will see a number of panels that provide you with quick access to information you need. You may view this information for each wave by clicking the Wave dropdown at the top of the Dashboard.

*Note: If you want to view the dashboard of another Service Area that you have access to, click **Switch Service Area** at the top right of your screen, under the current Service Area name.*

Route Summary – Lists the work areas for the day, with information on how many of each type of package is on each route, as well as route type, time, and distance.

ROUTE SUMMARY							
WORK AREA NAME	WA #	ROUTE TYPE	TIME	DISTANCE	LP STOPS	LP PKGS	BU STC
		REG	9.40	137	20	39	1
		BULK	1.59	68	0	0	1
		REG	5.68	89	15	25	3
		REG	7.31	91	13	17	0
		REG	10.76	151	20	38	2
		REG	13.44	129	20	31	2
		REG	9.08	78	14	36	6
		REG	9.62	118	19	25	0
		REG	4.24	82	5	6	1
		REG	7.78	97	9	10	3
Unassigned			0.00	0	135	227	1
Total:			78.90	1040	270	454	3

Routing Status – Provides information on today's routes. You may see the service area (number of stops and number of packages), the active route plan, the current solution, whether overtime is enabled or disabled (you may change the status here), and a solution comparison between the current solution and the informed optimal solution.

Routing Status

VOLUME: [Stops: 450](#) | [Packages: 832](#) | 100%

ROUTE PLAN: [Added New Route Plan](#)

SOLUTION: Not Locked **OVERTIME:** Disabled ☐ Enabled

SOLUTION	ROUTES	TIME	DIST	TOT.COST	DIFF
Actual	7	53.13	704.86	\$2,900.75	
Inf. Optimal	6	46.46	700.84	\$2,597.20	-10%

History – Shows historical data for the same day for the last four weeks, listing the number of routes, stops, and packages for that day each week.

Unrouteable Stops – Lists the stops for the Service Area that are unrouteable. A stop is unrouteable if the coordinates are not precise enough to be routed. For each stop, the table shows the number of packages, total weight, total cube, stop type and class, and address information for the stop. The **Eligible** tab lists the stops that are eligible for assignment.

UNROUTEABLE STOPS												
CSA	PACKAGES	TOTAL WEIGHT	TOTAL CUBE	STOP TYPE	STOP CLASS	FIRM NAME	ADDRESS	CITY	STATE	POSTAL CODE	WINDOW OPEN	WINDOW CLOSE
-1 301603 302241 302244 ...	2	1.3	2.05	Delivery	REG			CHESTER SPRINGS	PA	19425		
301603 302241 302246 90...	1	15.3	2.85	Delivery	REG			PORT WENTWORTH	GA	31407		
-1 301603 302241 302246 ...	6	120.9	19.69	Delivery	LP					19380		
301603 302241 302246 90...	2	0	5.66	Delivery	REG					19341		
301603	1	0	0.2	Delivery	REG			MORGANTOWN	PA	19543		
301603	1	0	5.92	Delivery	REG			GEIGERTOWN	PA	19523		

Map

Display Options

Dispatch – Enables you to choose the dispatch you want to see

Solution – You may choose to see the Actual solution or the Informed Optimal solution

Routes – a list of routes for the day in the current dispatch. Use the checkboxes to hide or show the routes on the map. The Stop Info table below the map will also show only the stops on the routes that you choose to view.

Map Tools

Search field – enables you to enter an address to find and zoom to on the map



Zoom In – Zooms the map in each time you click it



Zoom Out – Zooms the map out each time you click it



Default Map View – Returns the map to the default view



Pan – Click and hold on the map to drag the view to another area (image shows cursor on the map, not a button)



Zoom to Routes – Zooms the map to a view that includes all of your routes



Select Tool – Enables you to create a selection polygon



Clear Selection – Clears the current selection



Create Routes – Enables you to create routes using the active route plan during the planning window.



Geography Swap Tool - Enables you to use a selection polygon to select a group of stops on the map and swap them to another route temporarily (for the current sort date).



Swap Tool – Enabled after routes have been created and stops are selected in the table or on the map. The Swap tool enables you to move stops from one work area to another.



Lock Solution – Enables you to lock the current solution. Any stops that come into DRO after you lock the solution will be assigned based on the locked route structures. Bear in mind that locking a solution may limit dynamic route generation for future unassigned stops for the next day's sort.



Show Sequence Layer - Enables you to view the sequence layer for a route. You may view the sequence for one route at a time.



Show Anchor Areas - Enables you to view the current anchor areas on the map. You may show or hide anchor areas by choosing which routes are visible on the map.



Show postal layer - Displays postal code boundaries on the map.



Show relay layer - Displays relay locations on the map if relays are enabled for your station.

Optimization Summary - Enables you to view a summary of stops and routes that missed route constraints, including missing time critical stops, routes that exceeded capacity, routes that broke stop thresholds, and routes that exceeded route length.

Time Critical Stops - Displays time critical stops on the map by changing the symbols of those stops to a clock symbol.



Panel Toggle – Slide to hide or show the **Stop Information** panel below the map



Export – Enables you to export the map either plainly (Map Only) or with additional information like a title, author, copyright, and so on (Layout). You may export the map as a pdf or one of several image files.



Basemap – Enables you to choose a basemap to use on your map view

Waypoint Boundary Tool - Enables you to create, edit, and delete waypoint boundaries on the map.

Change Work Area Symbol - Enables you to change the shape and color for the work area symbols by work area. This can be either a temporary or permanent change.

Note: You may also use the mouse scroll wheel to zoom in and out on the map.

Stop Information Table Tools

The Stop Information table shows information about the stops on the visible routes on the map. Many of the Stop Information table tools become active when one or more stops is selected, either in the

table or on the map.

Show All dropdown - You may choose to display all stops on the visible routes by the type of stop they are. You can choose to see Pickup Stops, LP Stops (custom defaults), LP Stops (FedEx defaults), Bulk Stops, Small Stops, CDO Stops, or Time Critical Stops (All, On time, At risk, Missed, Unknown).

Search – Enables you to search the stops in the **Stop Information** table based on any field. As you enter your search term, the stops in the table will be restricted to just those fitting your search criteria.



Zoom to Selected Stops – When you have selected stops in the table or on the map, this tool zooms the map to an extent that includes all of the selected stops.



Show Selected Records – When you have selected stops in the table or on the map, this tool shows only the selected records in the **Stop Information** table



Show All Records – When the table is showing only selected records, this button will show all records again.



Create stop override - Enables you to create a stop override for each of the selected stops.

Toggle on/off Selected View - Displays only the selected stop(s) in the table and on the map.



Export to CSV - Enables you to download your **Stop Information** table in .csv format. The download will occur when you click the button.

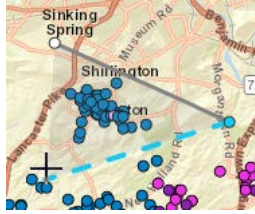
Map Functions

Selecting on the Map

You may use the **Select Tool** to easily select a group of stops on the map.



1. On the map page, click the **Select Tool** button.
2. Click once on the map to start drawing your selection polygon.
3. Continue clicking as needed to draw the polygon around the area you want to select.



4. When you are finished, double-click to create the last point. The stops within your polygon will be selected.

Creating Routes

During the planning window, you may create routes using the active route plan for the day for the current set of waypoints. After routes are created, the waypoints on the map will be classified by the assigned route value, and the Map and Report pages will be updated with information on the routes. In addition, Actual and Informed Optimal solutions are created.

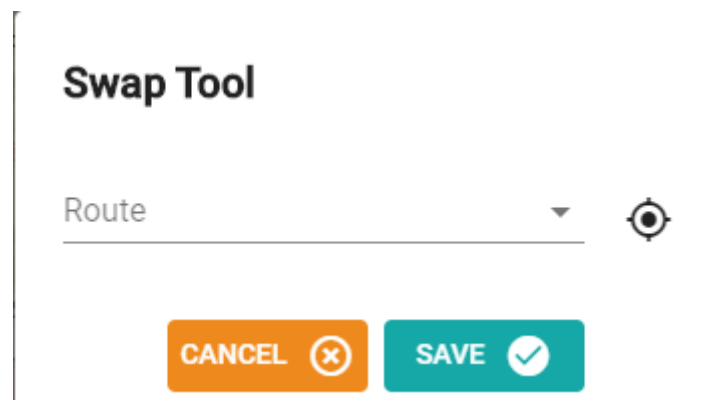


1. To create routes, click the **Create Routes** button on the map.
2. On the Create Routes window that opens, you can choose whether to create an Informed Optimal solution. Choose **Yes** or **No**.
3. Click **Confirm** to create routes.

Wait for the solver to be called. Resulting routes will be created using the scheduled route plan and the currently available stops. It can take up to 10-15 minutes for all routes to be created. Avoid hitting Create Routes again before this time.

Swapping Individual Waypoints

To fine-tune your routes, you may find it necessary to move some points from one route to another. The Swap Tool button on the Map page lets you easily swap waypoints between routes during the planning window.





1. Select the stop(s) you want to move to another route. You may select these stops on the map using the **Select Tool**, or you may select them in the Stop Info table below the map.
2. Click the **Swap Tool** button.
3. On the **Swap Tool** window that opens, click the **Route** dropdown to choose the route to which your stop(s) should be assigned.



*Note: If you do not know which route to choose, you may use the **Identify** tool and then click your target route on the map. The Route dropdown will reflect the route you clicked.*

4. After you have chosen your target route, click **Save**. Your stop(s) will be moved to the other route.
5. Click the **Swaps** tab in the Stop Information table below the map. The Swaps tab shows all of the individual and geographic swaps you have made. Swaps are temporary and are included in the next day's sort.

Using the Geography Swap Tool

The Geography Swap tool enables you to transfer areas of geography between routes on a temporary basis for the current sort date. Using this tool, you may not only swap visible stops within an area (as with the individual swap function), but also the geography itself, ensuring that future stops will also transfer. When you use this tool, you will be prompted to draw and assign geography, similar to how anchor areas are created. This geography will then serve as a temporary anchor area for the current sort date only, and will take precedence over any underlying geography assignment of the same route type. The geography swaps created on that sort date can also be viewed on the Geographic tab under Swaps on the Stop Info table for easier visibility and management.



1. Click the **Geography Swap tool** button.
2. On the **Geographic Swap Type** pop-up menu, choose the type of stops you are swapping.
3. Use the cursor on the map to click and draw a polygon around the stops you want to select.
4. Enter a name for this swap.
5. Click the **Route** dropdown and choose the route to which the selected stops should be moved.
6. Click **Save**. The swap will be displayed on the Geographic tab of the Swaps view in the Stop Info table.

Showing Anchor Areas



1. To view the anchor areas for the current routes, click the **Show anchor areas** button.
2. On the pop-up menu that opens, choose the type of anchor areas you want to see (REG, LP, Bulk, Small, Bulk/LP). Anchor areas will be displayed on the map for any visible routes that have anchor areas.

3. To show or hide anchor areas, use the checkboxes in the Routes list to show or hide routes on the map.
4. When you are finished, click the button again to hide the anchor areas.

Showing Postal Codes

Note: You may need to hide the Stop Information table to see all of the map tools.



Click the **Show postal code** button. Postal code boundaries will be displayed on the map. Click the button again to hide the postal code display.

Showing the Relay Layer

The relay layer contains the relay start and end points. The Station location will appear as a red flag icon, and available relay locations will appear as white flag icons. You may click a station or relay symbol to view information about the location.



Click the **Show relay layer** button to view the relay layer.

Viewing the Optimization Summary

The Optimization Summary window displays information for stops and routes that have exceeded certain thresholds.

1. Click the Optimization Summary button.
2. On the Optimization Summary window, use the tabs to view stops and/or routes that have exceeded the listed thresholds.
3. For each of the categories, you may use the Export to csv button to export the list of stops or routes.
4. Close the window when you are finished.

Displaying Time Critical Stops

A time-critical stop refers to any stop with a specific time window, such as a scheduled pickup, appointment delivery, or stop with a latest delivery time. This new time-critical functionality can be used to:

- Visually identify time-critical stops more easily on the map
- View detailed information regarding these stops
- Identify where the current routing solution suggests a time-critical stop may be missed

- Identify where the current routing solution suggests a time-critical stop may be delivered, but within 15 minutes of time window opening or closing ("At Risk")
- Identify specific routes that may need to be adjusted to accommodate time-critical stops
- Highlight various subcategories such as service in time, missed, "At Risk", or all time critical stops

To see on the map which stops are time critical, click the **Time Critical Stops** button. The symbols for those stops will be changed on the map to a clock symbol (on-time), exclamation point (missed), or horizontal line (at risk).

- **Missed** - indicates the routing solution suggests the stop will be missed
- **At Risk** - indicates the routing solution suggests the stop will be completed within 15 minutes of the window open or close
- **On-Time** - indicates the routing solution suggests the stop will be completed safely within range

Click the Time Critical Stops tab in the Stop Information table below the map. On this tab, you will see the list of all time critical stops, as well as stop and delivery information for each. You may use the Show All dropdown at the top of the table to filter the records in the table to each of the time critical stop categories.

Showing the Sequence Layer

1. Use the routes list on the left side of the screen to display only one route on the map. You may view sequence information for only one route at a time.
2. Click the **View sequence layer** button. The sequence layer for the route will be displayed on the map.

Exporting Your Map

The Export feature enables you to save your map as a pdf or image file.



Click the **Export** button to export your map view.

Exporting Layout

1. On the **Layout** tab, enter the title you want to use for your map.
2. Click the **Page Setup** dropdown to choose the page size and orientation (**Portrait** or **Landscape**) you want to export.
3. Choose the file format you want to receive. You may choose **PDF** or one of several image types.

4. Click **Advanced Options** if you want to set the map scale, or add **Author** or **Copyright** information to the output. You may also choose the **DPI** at which the output file would print, and whether you want to include a legend and north arrow.
5. When you are finished, click **Export**. When the file has been created, it will appear below the export button, and you will be able to download it from there.

Exporting Map Only

1. On the **Map Only** tab, enter the file name you want to use for the export file.
2. Choose the file format you want to export. You may choose **PDF** or one of several image types.
3. Choose the **Width** and **Height** of your output map.
4. Click **Advanced Options** if you want to adjust the map scale, the **DPI** at which your export file would print, or whether you want to include attribution on the map.
5. When you are finished, click **Export**. When the file has been created, it will appear below the export button, and you will be able to download it from there.

Changing a Work Area Symbol

The Change Work Area Symbol tool enables you to change the color and/or symbol of work areas on your map. The change can be temporary or permanent.

1. Click the Change Work Area Symbol button.
2. On the window that opens, click the Work Area dropdown to choose the work area you want to change. The color and symbol will be updated to display the current assignment.
3. Click the color box to choose a new color for the work area. You may use the color picker, or you may enter a color definition, if you have one.
4. Click the Shape dropdown to choose a new shape to represent the stops on the map. You may choose from circle, square, diamond, or triangle.
5. If you want to make this change permanent, click the checkbox next to *Make this the new default symbol for this WA*.
6. When you are finished, click **Save**.

Reports

The Report page shows you reports for your routes. At the top of the page you will see the current CSA, Station, and Date. Click the Actual or Informed Optimal radio buttons to choose which reports you want to see.

Click the tab for a report in the table to review the report.

Solution Summary – The Solution Summary report shows projections for Actual and Informed Optimal solutions, including number of routes, total time, total distance, fixed and variable costs for both vehicles and drivers, overtime, and total cost, as well as a summary of the differences between the Actual and Informed Optimal solutions. As with any projection, actual results may vary and should be evaluated in light of the service provider's independent business judgment and knowledge of its Service Area.

Note: The default values for Fixed and Variable Vehicle costs are based on typical vehicle price, mileage, wear-and-tear, maintenance, fuel, and other factors. However, actual vehicle costs will vary with each business and Service Area. Accordingly, to improve the helpfulness of the Actual and Informed Optimal solution projections, service providers may wish to adjust the Fixed and Variable costs to reflect the costs their business is actually incurring. This can be done in the Defaults page of DRO. Fixed and Variable Driver Costs have no default value and are set by the service provider in the Fleet page of DRO.

Package Detail – The Package Detail report shows the work areas in the solution. For each work area, the report shows the route type, time, distance, delivery information (stops, packages, cube capacity vs available capacity, TW Miss), pickup information (stops, TW Miss), and Combo information (Stops, Packages, Cube, TW Miss). At the bottom of the report there is a total for each field. Note that the capacity field will be highlighted if the calculated capacity exceeds the listed vehicle capacity.

Cost Detail – The Cost Detail report shows the work areas and their associated route type in the solution. For each work area, projections for time, and distance, as well as the Vehicle cost (fixed and variable), the Driver cost (Fixed, Variable, and Overtime), and the total cost. At the bottom of the report there is a total for each field.

Note: The default values for Fixed and Variable Vehicle Costs are based on typical vehicle price, mileage, wear-and-tear, maintenance, fuel and other factors. However, actual vehicle costs will vary with each business and Service Area. Accordingly, to improve the helpfulness of the Actual and Informed Optimal solution projections, service providers may wish to adjust the Fixed and Variable Vehicle Costs to reflect the costs their business is actually incurring. This can be done in the Defaults page of DRO. Fixed and Variable Driver Costs have no default value, and are set by the service provider in the Fleet page of DRO.

Time and Distance Detail – The Time and Distance Detail report shows the work areas in the solution. For each work area the report shows the route type, time components (total, time, stem out, stem back, in zone, waiting, at stop, break), and distance components (distance, stem out, stem back, in zone). At the bottom of the report there is a total for each field.

Package Type Detail – The Package Type Detail report shows the work areas in the solution. For each work area the report shows the route type, total time, distance, and package type summary: LP stops, LP packages, Bulk stops, Bulk packages, Regular stops, and Regular packages. At the bottom of the report there is a total for each field.

Section Availability - The Section Availability report shows basic work area information (work area name, work area number, route type), summary statistics (cube or pkg per section, projected generic SIDs), as well as a summary of disabled sections, if applicable.

This report can be utilized by FedEx Ground station staff to communicate any Section Availability changes to FedEx Ground preload staff.

Inbound Package - The Inbound Package report shows you how many inbound LP, Bulk, CDO, or Small packages are at a stop. For each stop with LP, Bulk, CDO, or Small packages, the report also shows information about the stop (name and address information, sequence number, time window details) and the packages (total weight, total cube). You can choose to see LP, Bulk, CDO, or Small packages on all routes, or choose a particular route to view.

Stop Assignment Summary – The Stop Assignment Summary report shows, for each route, details about stops that were dynamically assigned versus directly assigned. It also shows the information for the stops that were manually adjusted for each route.

Dynamic – This assignment type occurs when a stop’s work area assignment is not directly predetermined by the user, but rather by the DRO solver based on that day’s volume and user-defined requirements, such as available drivers and vehicles, driver hours, and vehicle capacity. This type of assignment can occur via:

- **Dynamic Stop** – A stop with no defined constraint of any sort which is eligible to be assigned to the most optimal work area in a user’s available fleet for that day
- **Dynamic Anchor Area** – A user defined geographic area where stops are optimally distributed only among select, eligible work areas, as defined by the user.

Direct – This assignment type occurs when a stop is given a predetermined, single work area assignment by the user prior to optimization. A direct assignment can occur through use of:

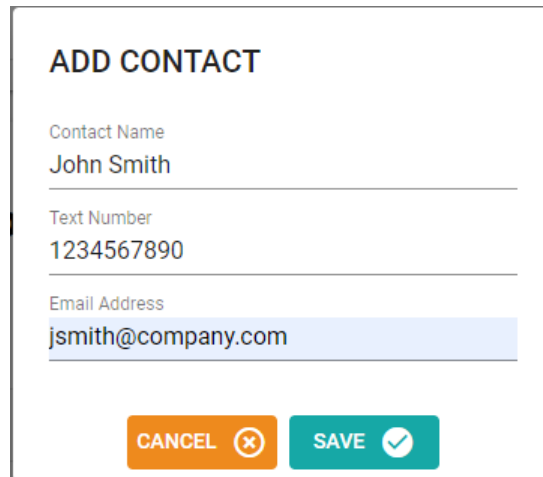
- **Anchor Area** – A user defined geographic area attached to a specific route
- **Stop Override** – A stop location assigned to a specific route.

Manual Adjustment – A stop that is manually assigned by the user to a different route than was originally determined following an optimization. This can occur through use of:

- **Geography Swap** – A user defined geographic area assigned to a different work area than originally intended, for the current sort date only
- **Individual Swap** – A stop location reassigned to a different work area than was originally intended, for the current sort date only
- **Unrouteable Assignment** – A stop that is unable to be assigned and sequenced during optimization due to partial address information, but is manually assigned to a work area by the user

Managing Communications

Adding Contacts to the Global Contact List



ADD CONTACT

Contact Name
John Smith

Text Number
1234567890

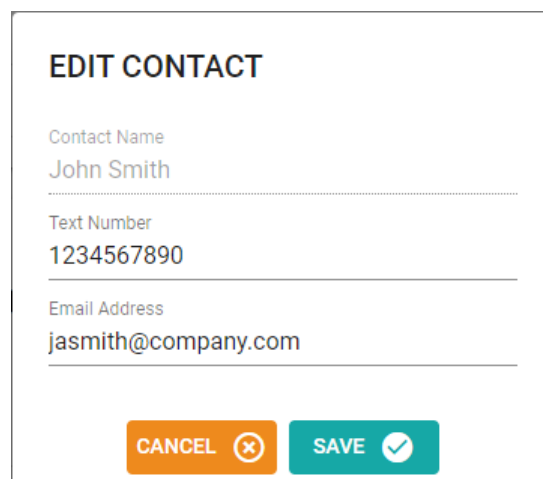
Email Address
jsmith@company.com

CANCEL ✕ SAVE ✓

ADD +

1. To add a contact to the Global Contact List, click **Add** in the top section on the Communications page.
2. On the **Add Contact** window that opens, enter the contact name, text number, and email address.
3. Click **Save**. The contact will be added to your list of contacts.

Editing a Contact



EDIT CONTACT

Contact Name
John Smith

Text Number
1234567890

Email Address
jsmith@company.com

CANCEL ✕ SAVE ✓



1. To edit a contact on the Global Contact List, hover over the contact you want to change and click **Edit**.
2. On the **Edit Contact** window that opens, make the changes you want to make to the contact's information. **Contact Name** is not editable.
3. When you are finished, click **Save** to save the changes.

Deleting a Contact

DELETE CONTACT

Are you sure you want to delete contact **John Smith** ?

NO YES 

1. To delete a contact from the Global Contact List, hover over the contact you want to delete and click **Delete**.
2. You will be asked to confirm that you want to delete the selected contact. Click **Yes** to delete the contact or click **No** to cancel.

Editing Alerts

Expected Volume exceeds current plan capacity

Contact	Text	Email
Julie	☐	☐
Ray	☐	☐

CANCEL SAVE 

A teal rectangular button with the word "EDIT" in white capital letters and a white pencil icon to its right.

1. To edit the contacts who receive an alert, hover over the alert event you want to edit and click **Edit**.
2. On the <**Event Alert Name**> window that opens, click the checkboxes for **Text** and/or **Email** for each contact that should receive the event alert.
3. When you are finished, click **Save**. The contacts for each alert will be updated on the main **Communications** page.

Managing Your Fleet

In the **Driver Management** section on the **Fleet Management** page, you may add, edit, and delete drivers. You may also use the switch to indicate whether overtime is enabled or disabled.

Adding a Driver

A Driver may be added to your fleet to manage the route they are associated with and to apply their specific wage rates to the calculated route cost, when optimized.

ADD DRIVER

Driver Name *

Joe Smith

Driver ID

24601

REGULAR

5

Fixed Driver Cost Annually

\$ 0.00

Driver Wage Hourly

\$ 0.00

Overtime Wage Hourly

\$ 0.00

Overtime Threshold: Hours/Day

8

CANCEL

SAVE

1. Verify you are on the Driver Management tab.
2. Click **Add** to add a driver.
3. On the **Add Driver** window, enter the driver name (required).
4. Enter the **Driver ID** (optional, for reference only).

5. Click the **Vehicle** dropdown and choose the vehicle for the driver (required). A vehicle must already be created to populate (grayed-out vehicles already have a driver assigned to them).
6. Enter the costs associated with the driver: fixed driver cost annually, driver wage hourly, and overtime wage hourly. These fields are optional.
7. Enter the number of hours per day after which overtime starts (optional). The previously entered Overtime Wage will be applied to any route cost calculation utilizing this driver where this Overtime Threshold is exceeded. Overtime rules and regulations vary by jurisdiction; service providers may wish to consult with independent legal counsel regarding specific obligations in their jurisdiction.
8. Click **Save**. Your driver will appear in the driver list on the Fleet Management page.

Editing a Driver

EDIT DRIVER

Driver Name *

Joe Smith

Driver ID

24601

REGULAR

▼

5

▼

Fixed Driver Cost Annually

\$ 0

Driver Wage Hourly

\$ 0

Overtime Wage Hourly

\$ 0

Overtime Threshold: Hours/Day

8

CANCEL ✕

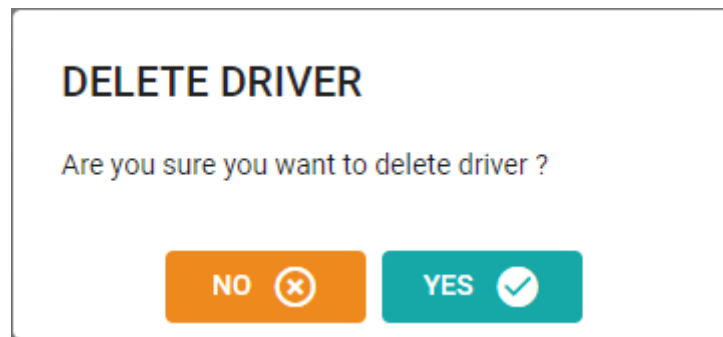
SAVE ✓

1. Verify you are on the Driver Management tab.



2. To edit a driver, hover over the driver you want to change and click **Edit**.
3. On the **Edit Driver** window that opens, make the changes you want to the driver information.
4. When you are finished, click **Save**. The updated driver information will appear on the **Driver** list on the **Fleet** page.

Deleting a Driver



1. Verify you are on the Driver Management tab.
2. To delete a driver, hover over the driver in the list and click **Delete**.
3. When you are asked to confirm the deletion, click **Yes** to delete the driver.

Adding a Vehicle

ADD VEHICLE ×

Work Area

Vehicle ID

123456

Vehicle Type

P900

Vehicle Capacity

540

Route Type

REG

☒ Front to Back Assignment

Load Profile

Section

Select Template (Optional)

Description: This loading profile will attempt to balance stops across the sections location in your van.

Examples: 1001, 1052, 1098, 1501, ...

Contingency: Packages will be assigned to sections of your van (e.g. 1000, 1500, etc).

SECTION AVAIL.

CANCEL

SAVE



1. Verify you are on the Vehicle Management tab.
2. Click **Add** to add a vehicle.
3. On the **Add Vehicle** window, click the **Work Area Name** dropdown and choose a work area. The Work Area names are populated from the Work Area Master Table, along with the current Work Area Number. If a work area is not available, it is either already associated with a vehicle, or will need to be added by Station Management.

*Note: **Work Area** is required.*

4. Enter a **Vehicle ID** for the vehicle you are adding. Vehicle ID can be alpha and/or numeric. This ID is for CSP reference only.
5. Click the **Vehicle Type** dropdown and choose the type of vehicle. If the Vehicle Type is not listed, it will need to be added in the Details section under Vehicle Cost Profile.

6. **Vehicle Capacity** is automatically populated when you choose the **Vehicle Type**, but you may change it if needed. This should be the actual, demonstrated capacity within the vehicle, which by default is 60% of actual vehicle cube. If not otherwise dictated through Anchor Areas or Stop Overrides, a vehicle will not be loaded beyond this stated capacity.
7. Click the **Route Type** dropdown and choose the type of route this vehicle services.
 - **Bulk Routes** - Bulk route rules can be defined in the Defaults tab. A stop will be considered a bulk stop and assigned to this route type if it exceeds either the package minimum or cubic foot minimum that is defined under **Bulk Stop Customization** in the Defaults tab of DRO.
 - **LP Routes** - Large Package routes will only be eligible to service packages specifically as LP. Additional preferences for LP routes can be configured under **Large Package Customization** in Defaults.
 - **REG Routes** - Regular routes do not consider user-defined route type rules.
 - **SM Routes** - Small routes can be defined in the Defaults tab. a stop will be considered a Small stop if it meets the criteria defined under **Small Stop Customization**.
8. **Front to Back Assignment** indicates whether the cargo area can be accessed through the front of the vehicle. If selected, packages will be loaded with the first stops in sequence toward the front of the vehicle and last stops toward the back, unless otherwise dictated through a Stop Override. If not selected, packages will be loaded with the last stops in sequence toward the front of the vehicle and the first stops toward the back, unless otherwise dictated through a Stop Override.
9. Click the **Load Profile** dropdown and choose a load profile. The load profile can be modified here from the default selection of **Section** loading. Please note that when **No Section** is selected, packages will not receive shelf assignments and will indicate to the loader to stage packages, as opposed to van loading.

Note: You will see a description of the Load Profile you chose below the dropdown.

10. If you choose Section as your load profile, the **Section Avail** button will become available. Section availability enables you to customize the section configuration for the vehicle. See **Customizing Section Availability** below for more information.
11. Choose a color and shape that will be used when displaying this work area on the map. You can use the color picker or use color definitions.
12. Click **Save** when you are finished.

Note: If you want the vehicle you added to be used by an existing route plan, remember to add it to any plan it needs to be on.

Customizing Section Availability

ADD VEHICLE - SECTION AVAIL.

Stop overrides to a section that is turned off will still be respected.

☒ Select All

Section	Weight	Availability
Left 1	3000	☒
Left 2	3500	☒
Left 3	7000	☒
Left 4	7500	☒
Left 5	4000	☒
Left 6	4500	☒
Left 7	8000	☒
Left 8	8500	☒
Left 9	3xxxFL	☒
Left 10	4xxxFL	☒
Left 11	7xxxFL	☒
Left 12	8xxxFL	☒
Center 1		
Center 2		
Center 3		
Right 1	1000	☒
Right 2	1500	☒
Right 3	5000	☒
Right 4	5500	☒
Right 5	2000	☒
Right 6	2500	☒
Right 7	6000	☒
Right 8	6500	☒
Right 9	1xxxFL	☒
Right 10	2xxxFL	☒
Right 11	5xxxFL	☒
Right 12	6xxxFL	☒

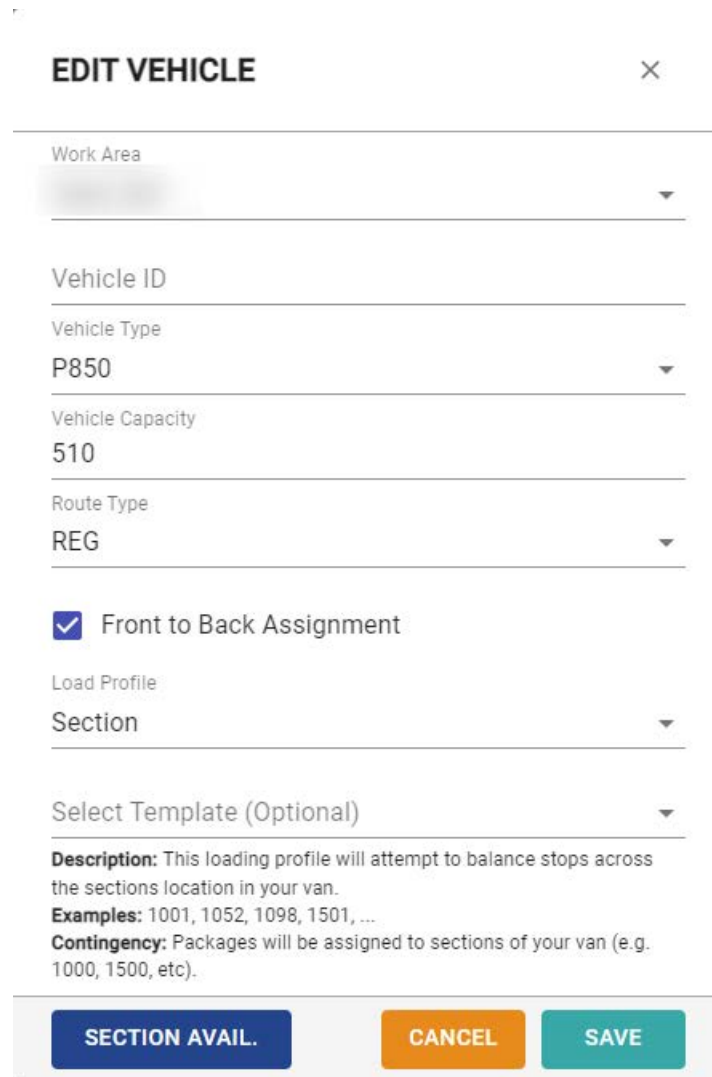
When you add, copy, or edit a vehicle and choose **Section** as the load profile, you will have the ability to customize the section availability of the vehicle you are setting up.

Section Availability provides you the option to disable shelf and floor sections from the standard 16 section cargo area configuration. DRO will then recognize disabled sections when generating Section ID's (SIDs) and distribute volume as evenly as possible across only available sections of the vehicle, as designated by you. This customization will be specific to the vehicle, and will be utilized in each route plan for which the vehicle is active.

Since modifications are unique to the vehicle, if different section configurations for this vehicle are intended for different days of the week, the Section Availability configuration for this vehicle must be edited prior to submitting the route plan.

1. On the Add, Copy or Edit Vehicle window, click **Section Avail.**
2. Use the checkboxes to indicate the sections that are available or unavailable in the vehicle.
3. Click **Save**.

Editing a Vehicle



EDIT VEHICLE ✕

Work Area

Vehicle ID

Vehicle Type

P850

Vehicle Capacity

510

Route Type

REG

☒ Front to Back Assignment

Load Profile

Section

Select Template (Optional)

Description: This loading profile will attempt to balance stops across the sections location in your van.

Examples: 1001, 1052, 1098, 1501, ...

Contingency: Packages will be assigned to sections of your van (e.g. 1000, 1500, etc).

SECTION AVAIL. **CANCEL** **SAVE**

1. Verify you are on the Vehicle Management tab.
2. To edit a vehicle, hover over the vehicle you want to change and click **Edit**.
3. On the **Edit Vehicle** window that opens, make the changes you want to the vehicle information.
4. When you are finished, click **Save**. The updated vehicle information will appear in the **Vehicle** list on the **Fleet** page.

EDIT 

Copying a Vehicle

COPY VEHICLE ×

Work Area

Vehicle ID

309234

Vehicle Type

P500

Vehicle Capacity

300

Route Type

REG

☒ Front to Back Assignment

Load Profile

Section

Select Template (Optional)

Description: This loading profile will attempt to balance stops across the sections location in your van.
Examples: 1001, 1052, 1098, 1501, ...
Contingency: Packages will be assigned to sections of your van (e.g. 1000, 1500, etc).

SECTION AVAIL.

CANCEL

SAVE

You may want to copy a vehicle to easily add another vehicle of the same type/details without having to re-enter all of the vehicle information.

1. Verify you are on the Vehicle Management tab.
2. To copy a vehicle, hover over the vehicle you want to copy and click **Copy**.
3. On the **Copy Vehicle** window that opens, make the changes that you want to the new vehicle and click **Save**.

Note: Work Area Name is a required field.

Deleting a Vehicle

DELETE VEHICLE

Warning: This vehicle is used in one or more route plans.

Are you sure you want to delete vehicle 6 ?

NO YES

1. Verify you are on the Vehicle Management tab.
2. To delete a vehicle, hover over the vehicle in the list and click **Delete**.
3. If the vehicle is assigned to a route plan you will be warned. Click **Yes** to confirm that you want to delete the vehicle.



Adding a Vehicle Cost Profile

SELECT THE TYPES OF VEHICLES IN YOUR FLEET:

AVAILABLE

P850

P550

P500

P400

P350

P300

>

<

SELECTED

P700

P750

P650

P600

P450

P900

CANCEL SAVE

The Vehicle Cost Profiles section on the Fleet page lists all of the vehicle types in your fleet. You should ensure that this section includes all of your vehicle types because they appear as options when adding or editing individual vehicles in your fleet.

1. Verify that you are on the Vehicle Cost Profiles tab.



2. To add a vehicle cost profile, click **Add**.

3. On the window that opens, click the vehicle types you want to add and use the arrows to move them from the **Available** field to the **Selected** field.
4. When you are finished, click **Save**. Your vehicles will be added to the list of vehicles in the **Vehicle Cost Profiles** section, and from there you may edit the cost profile for each.

Editing a Vehicle Cost Profile

COST PROFILE: ST

Fixed Cost: \$ 10000 annually

Variable Cost: \$ 23 per mile

CANCEL (X) SAVE (checkmark)

1. Verify that you are on the Vehicle Cost Profiles tab.
2. To edit a vehicle cost profile, hover over the vehicle you want to edit and click **Edit**.
3. On the **Cost Profile: <Vehicle Type>** window that opens, make changes to the **Fixed Cost** and/or **Variable Cost** for the vehicle type.
4. Click **Save** when you are finished.

EDIT 

Deleting a Vehicle Cost Profile

DELETE VEHICLE

Are you sure you want to delete vehicle **ST** ?

NO (X) YES (checkmark)

1. Verify that you are on the Vehicle Cost Profiles tab.
2. To delete a vehicle cost profile, hover over the vehicle you want to delete and click **Delete**.

DELETE 

- When you are asked to confirm that you want to delete the vehicle, click **Yes** to delete, or **No** to return to the **Vehicle Cost Profiles** list.

Adding a Custom Load Profile

ADD LOAD PROFILE

Name

Description

Stop overrides to a section that is turned off will still be respected.

☒ Select All

3000
☒

3500
☒

7000
☒

7500
☒

4000
☒

4500
☒

8000
☒

8500
☒

3xxxFL
☒

4xxxFL
☒

7xxxFL
☒

8xxxFL
☒

1000
☒

1500
☒

5000
☒

5500
☒

2000
☒

2500
☒

6000
☒

6500
☒

1xxxFL
☒

2xxxFL
☒

5xxxFL
☒

6xxxFL
☒

CANCEL

SAVE

- Verify that you are on the Vehicle Cost Profiles tab.
- To add a custom load profile, click **Add**.



3. On the **Add Load Profile** window that opens, enter a name and description for the load profile you are creating.
4. Use the checkboxes to enable or disable the sections of the vehicle as needed.
5. When you are finished, click **Save**. Your load profile will be listed.

Editing a Custom Load Profile

1. Verify that you are on the Vehicle Cost Profiles tab.
2. To edit a custom load profile, hover over the profile you want to edit and click **Edit**.
3. On the **Edit Load Profile** window that opens, make the needed changes to your load profile.
4. When you are finished, click **Save**.

A teal rectangular button with the word "EDIT" in white capital letters and a white pencil icon to its right.

Deleting a Custom Load Profile

1. Verify that you are on the Vehicle Cost Profiles tab.
2. To delete a custom load profile, hover over the profile you want to delete and click **Delete**.
3. When you are asked to confirm that you want to delete the profile, click **Yes** to delete, or **No** to return to the **Custom Load Profiles** list.

A teal rectangular button with the word "DELETE" in white capital letters and a white trash can icon to its right.

Managing Route Plans

Creating a Route Plan

CREATE ROUTE PLAN

Route Plan Name

SELECT REGULAR ROUTES:

AVAILABLE ☐ Select All

☐ ☐ 10 ☐ 11 ☐ 14

SELECTED ☐ Select All

SELECT BULK ROUTES:

AVAILABLE ☐ Select All

SELECTED ☐ Select All

SELECT LARGE PACKAGE ROUTES:

CANCEL **SAVE**

A route plan is a template that can be created and saved for future use. It contains the unique combination of route parameters to be used on a given day (vehicles, anchor areas, stop overrides).

ADD

1. To create a **Route Plan**, click **Add** in the **Create** section on the **Route Plan** page.
2. On the **Create Route Plan** window that opens, enter a name for your route plan.
3. In each of the sections (Regular Routes, Bulk Routes, Large Package Routes, Small Routes), select the truck types to include in your plan by clicking the route and using the arrows to move them to the **Selected** lists.
4. When you are finished choosing your routes, click **Save**.
5. Your route plan will be added to the list of plans on the **Route Plan** page.

Editing a Route Plan

Route Plan: New route plan

Optimize On:

☒ Full Fleet

☐ Target Time

STOP OVERRIDES

NEIGHBORHOOD CLUSTERS

ADD/REMOVE VEHICLES

Order*	Wave	Work Area Name	Work Area Number	Vehicle ID	Route Type	Driver	Start Time	Route Length	Anchor Areas
0	1	2	0377	test	BULK		08:30	08:00	
1	1	6	0462	023	LP		08:30	08:00	
2	1	3	0131		REG		08:30	08:00	

*Values outside 0-1000 and non-unique values will be auto-assigned

CLOSE

Note: If you have created anchor areas and/or stop overrides, you may add them to a route plan here.

EDIT

1. To edit a route plan, hover over the plan you want to edit and click **Edit**.
2. On the route plan window that opens, you will see your active route plan. You may choose to optimize on **Full Fleet** or **Target Time**.
 - The **Full Fleet** option will balance workload across all routes in your route plan (while still respecting your anchor areas and stop overrides).
 - The **Target Time** option will assign work in the most optimal way using the least number of routes while respecting route length limits, route capacity, stop overrides, and anchor areas. Routes will be assigned work based on their priority in the **Order** column (routes with the lowest order will be utilized first when assigning work).
3. If you want to use Dynamic Anchor Areas for the route plan, check the box next to Use Dynamic Anchor Areas.

Note: Enabling Dynamic Anchor Areas will update interface options for the route plan so that dynamic anchor areas can be used. It is recommended that you save a copy of the route plan before using dynamic anchor areas.

*Use the buttons on the top right of the window to edit the **Stop Overrides** or **Vehicles** for the route plan. (See below.)*

4. When you are finished editing your route plan, click **Close**.

Editing the Work Areas in the Route Plan

EDIT

1. To edit one of the work areas in your route plan, hover over the work area you want to edit and click **Edit**.
2. On the window that opens, edit settings for the work area:

- **Route Type** - Change the type of the route if needed. You will receive a warning message about the impact of changing route type.
- **Dispatch** - Choose the dispatch you want to assign the work area to.
- **Helpers** - Service Providers who use helpers on their route(s) will have the ability to define the number of helpers per route. When helpers are designated, the service time applied to stops on the route will be scaled according to the number of helpers specified.
- **Start Time** - Enter the start time of the route. This is the expected dispatch time of the route.
- **Target Hours** - Enter the target hours for the route. This is the route length from dispatch to return to station.
- **Driver** - Choose the driver for the route. If the driver you want is not listed, you may add them on the Fleet page under Driver Management.
- **Stop Min** - Enables you to set a minimum stop count that you desire on a route
- **Stop Max** - Enables you to set a maximum stop count that you desire on a route

Note: For more information on Stop Min and Stop Max, see Using Stop Thresholds below.

- **Sequence Pickups** - When this setting is enabled (checked), DRO will sequence pickups and deliveries together and will attempt to meet all time windows. When this setting is disabled (unchecked), DRO will place all pickups with time windows of less than 7 hours at the end of the route.
- **End Route at Last Stop** - Ends the route at the last stop, rather than returning to a facility
- **Manual Assignments Only** - This setting may be used to restrict a specific route from dynamic assignment within an otherwise dynamic plan, or enable unique operating scenarios such as pickup only, overflow, swap only, or dedicated customer/location routes. This option also replaces the current workflow commonly used to restrict dynamic assignment of artificially low route capacity with anchor area, stop override, or swap assignments. Routes using this option will be labeled with a lock icon in a new column in the Route Plan screen, as well as on the map and dashboard pages.

Work Area Name	Work Area Number	Vehicle id	Driver	Sequence Pickups ☒	End Route At Last Stop ☐	Manual Assignments Only ☐
Route Type REG	Dispatch 1	Helpers 0	Start Time 08:30	Target Hours 08:00		

Anchor Areas

AVAILABLE

UNSPECIFIED

FIRST

LAST

SEQUENCED (5)

➡

⬅

CANCEL

SAVE

On the left side of the window you will see the list of anchor areas that are available for prioritization and sequencing. On the right side of the screen you will see the priority categories to which you may assign your anchor areas. (For more information on prioritizing and sequencing anchor areas, see **Anchor Area Sequencing and Prioritizing on page 44**).

3. To assign available anchor areas, click the category you want to assign anchor areas to.
4. Click an anchor area you want to move to the selected category.
5. Use the arrows to move the available anchor areas to the correct categories. You may also drag and drop from one side to the other.
6. If you have added anchor areas to the **Sequenced** category, drag them into the sequence order you want.
7. On LP routes, for each assigned anchor area, you may choose whether all stops in the anchor area will be serviced by the route, or only the LP stops in the anchor area will be serviced by the route. Click the button to toggle between All Stops and LP stops.
8. For Bulk routes, you can toggle between Bulk, Reg, and Bulk/LP
 - **Bulk** - the route will service only stops meeting the bulk criteria
 - **Bulk/LP** - the vehicle can service both Bulk and LP stops. This type of anchor area can be used only with Bulk routes.
9. When you are finished click **Save**.

Using Stop Thresholds

The Stop Min and Stop Max settings on the work area editing window enable you to specify a stop target for a route that the DRO solver will then attempt to meet, given existing user constraints. There are three methods for specifying a stop target:

- **Minimum Stop Threshold only** - Solver will attempt to create a route at or above a minimum specified stop count
- **Maximum Stop Threshold only** - Solver will attempt to create a route at or below a maximum specified stop count
- **Stop Range** - Solver will attempt to create a route within a specified minimum and maximum stop count

Note: Several constraining factors may lead to the stop thresholds not being adhered to, which include:

- If the inclusion of existing anchor area assignments causes the route to exceed the specified stop thresholds
- If the inclusion of existing stop overrides causes the route to exceed the specified stop thresholds
- If the number of swaps causes the route to exceed the specified stop thresholds
- If the resulting solution would cause a route to exceed available vehicle capacity, capacity will take priority
- In rare circumstances where there is not available fleet to reasonably meet all stop demands
- If a solution is not possible given existing route constraints, solver will approximate as close to parameters as possible
- Stop Swapping
 - If a user optimizes using stop thresholds and then swaps stops such that the thresholds are then no longer met, the DRO solver will still attempt to achieve the stop thresholds during Final Optimization
 - For example, if a user creates a 100-stop route based on a specified a 100-stop minimum threshold, and then swaps 10 of those stops to another route, when Final Optimization is run, the DRO solver will respect the swap assignments, but attempt to supplement the route with 10 additional stops to achieve the 100-stop minimum
- Manual Assignment Routes - If a route is designated as 'Manual Assignment Only,' any existing route stop thresholds will not be adhered to as dynamic assignments are needed to achieve specified stop parameters

Managing Active Stop Overrides

Filter

Enabled	Stop Name	Stop Address	Postal Code	Stop Type	Route Override	Stop Order	Section	Location
☐	TYLER UNGER	129 S CHURCH ST		Both	Bulk	Last	4000	FL
☐	TYLER KNARR	83 WENTZEL RD		Delivery	LP	Last	1500	
☐	LEAVE AT FRONT DOOR	584 WOLFE LN		Both	N/A	Last		
☐	DAKOTA SOKOL	521 GROVE AVE		Both	N/A	Last		
☐	DAKOTA SOKOL	521 GROVE AVE		Both	N/A	First		
☐	GARY K DOERRIES	3 CHESTNUT HILL DR		Both	N/A	First		
☐	DOUGLAS RIGG	770 FAIRMONT AVE		Both	N/A	No Preference	9000	FL
☐	JOSHUA TYLER RADWANSKI	207 RUDLOFF LN		Delivery	SNR 4	First	9000	FL
☐	DAN THE TIRE MAN	3371 MORGANTOWN RD		Both	N/A	First		
☐	RALPH DELILLO	44 SHEA DR		Both	N/A	Last		
☐	JULIETTE FONTE	21 CINAMON CT		Both	N/A	First		
☐	KAREN ROYER	641 KURTZ MILL RD		Both	N/A	First		
☐	PAT FISHER	10 S GRANDVIEW AVE		Both	N/A	Last		
☐	BEEM URSULA	458 LEDGE ROCK RD		Both	N/A	First		
☐	AJ677086)	5006 HIGHLAND DR		Both	N/A	No Preference	1000	
☐	TYLER KNARR	83 WENTZEL RD		Pickup	LP	First		

CANCEL  SAVE 

1. To edit the **Stop Overrides** for your route plan, click the **Stop Overrides** button at the top right of the **Edit Route Plan** window.
2. Choose which stop overrides should be used in the active route plan.
3. Click **Save** when you are finished.

Adding/Removing Vehicles

The screenshot shows a web interface titled "CREATE ROUTE PLAN". At the top, there is a "Route Plan Name" field with the text "New route plan". Below this, there are two sections for selecting routes. The first section, "SELECT BULK ROUTES:", contains two columns: "AVAILABLE" and "SELECTED". The "AVAILABLE" column lists three items: "SUPP1", "SUPP2", and "SUPP4", each with a checkbox. The "SELECTED" column shows one item: "2", also with a checkbox. Between these columns are right-pointing and left-pointing arrows. The second section, "SELECT LARGE PACKAGE ROUTES:", also has "AVAILABLE" and "SELECTED" columns. The "AVAILABLE" column shows one item: "6", with a checkbox. The "SELECTED" column is empty. At the bottom of the window, there are two buttons: "CANCEL" (orange) and "SAVE" (teal).

1. To edit the vehicles for your route plan, click the **Add/Remove Vehicles** button at the top right of the **Edit Route Plan** window.
2. You will see the chosen vehicles for your active route plan. Use the arrows to remove any vehicles necessary.
3. Click **Save** when you are finished.

Editing Relays in a Route Plan

If you have relays enabled on the **Defaults** page, you are able to edit relays in your route plans.

EDIT 

1. Hover over the route plan for which you want to edit relays and click **Edit**.
2. On the route plan window that opens, click **Relays** at the top left. The number in parentheses on the Relays button indicates the number of routes that have active relay locations assigned.
3. On the **Relays** window, you will see a list of your work areas with the work area number, relay start, and relay end for each.
4. Use the **Relay Start** and **Relay End** dropdowns to choose starting and ending locations for each work area.

Note: For any Relay Start or Relay End location that you leave blank, the route will use the station location.

5. Click **Save** when you are finished.

Editing Start and End Points by Route

If you have relays enabled on the **Defaults** page, you are able to choose the start and end point for each route.

EDIT 

1. Hover over the route plan you want to edit and click **Edit**.
2. On the route plan window, hover over the route you want to edit and click **Edit**. You may need to scroll the routes list to see the Edit button on the right side.
3. On the window that opens, use the **Relay Start** and **Relay End** dropdowns to choose the Start and/or End points for the route. If you do not choose a Start or End point, the route will use the station as the starting or ending location.
4. Click **Save** when you are finished.

Copying a Route Plan

COPY ROUTE PLAN

Copying: **Test Plan**

New Route Plan Name

CANCEL



SAVE



COPY 

1. To make a copy of one of your route plans, hover over the route plan you want to copy and click **Copy**.
2. On the Copy Route Plan dialog that opens, enter a name for your new route plan.
3. Click **Save**.

After your new route plan has been created, it will appear in the route plans list and from that point you may edit or schedule it as needed.

Deleting a Route Plan

DELETE ROUTE

Are you sure you want to delete **New route plan**

CANCEL SAVE 

Important Note: You will not be able to delete a route plan if it is currently assigned in your Route Plan schedule. You must remove it from the schedule before deleting.



1. To delete a route plan, hover over the plan you want to delete and click **Delete**.
2. When you are asked to confirm that you want to delete the route plan, click **Yes** to delete or **No** to return to the route plan list.

Scheduling Your Route Plans

Edit Route Plan: Monday

Route Plan Name

New route plan 

CANCEL SAVE 

Note: A route plan must be assigned to each day of the week, including Sunday.



1. To schedule your route plans, hover over the day of the week you want to schedule and click **Edit**.
2. On the **Edit Route Plan: <Day Name>** window that opens, click the **Route Plan Name** dropdown and choose the route plan you want to use for that day.
3. Click **Save**. The list will be updated with the route plan name and details.

Managing Anchor Areas

An Anchor Area is a user-created polygon on the map that is used to group stops. These polygons can then be attached to routes to specify that they must be completed by a specific route or serviced together as part of any solution. For example, you may want an anchor area around a gated community so that the community is on the same route, or you may want to force a retail center or shopping mall to be serviced by a particular LP route.

*Tip: Use **Find** to zoom to a particular address or place.*

Adding an Anchor Area

ADD ANCHOR AREA

Anchor Area Name

Southwest

Label

Southwestern area

CANCEL 

SAVE 

1. If you want to see anchor areas assigned to only one route plan, click the Route Plan dropdown and choose the one you want to view. If you choose a route plan, you will see the list of routes in the plan and you may choose which routes you want to see on the map.



2. To add an anchor area, click **Add** on the Anchor Area page.

Note: You may want to use the toggle at the bottom left of the map to hide the Anchor Areas table temporarily and see a larger map.

3. Click on the map to draw a polygon representing the anchor area you want to create. To create the polygon, click once for each point, and then double-click to end the polygon.



4. When you are finished adding the anchor area on the map, the **Add Anchor Area** window will open. Enter a name and label for your Anchor Area. Click **Save**.

Note: Creating overlapping anchor areas can have a unexpected impact on the optimized solution.

When you are finished adding your anchor areas, use the **Edit Route Plan** tool on the **Route Plans** management page to add it to the route plan(s) for which it should be used.

Renaming Anchor Areas

RENAME ANCHOR AREA

Anchor Area Name

Southwest 1246809

Label

Southwestern area

CANCEL 
SAVE 



1. To rename an anchor area, hover over the anchor area in the list below the map and click **Edit**.
2. Make changes to the anchor area name and/or label.
3. Click **Save** when you are finished.

Reshaping an Anchor Area

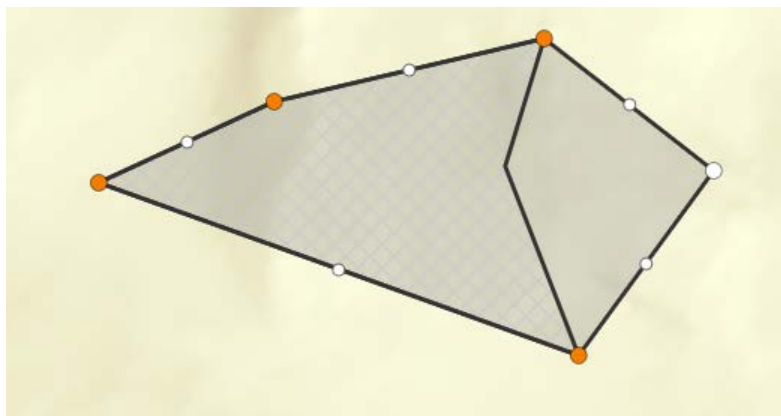
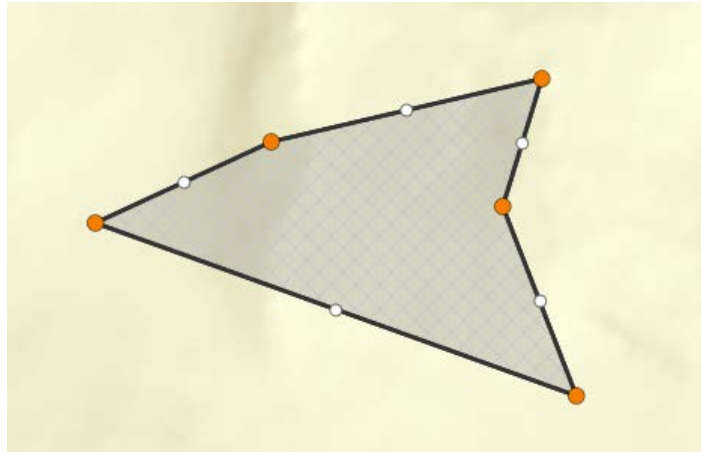
To reshape an anchor area, zoom in on the map to the anchor area you want to reshape.

If you want to see anchor areas assigned to only one route plan, click the Route Plan dropdown and choose the one you want to view. If you choose a route plan, you will see the list of routes in the plan and you may choose which routes you want to see on the map.

Method 1



1. Hover over the anchor area you want to change and click **Reshape**.
2. Use the **Orange** anchor points to move or resize connected line segments.

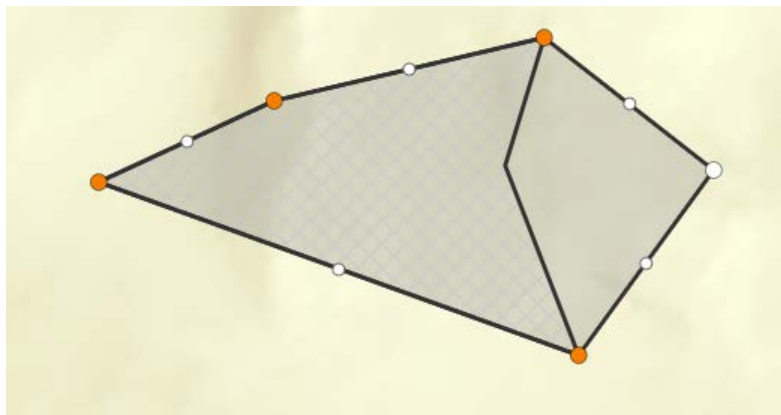
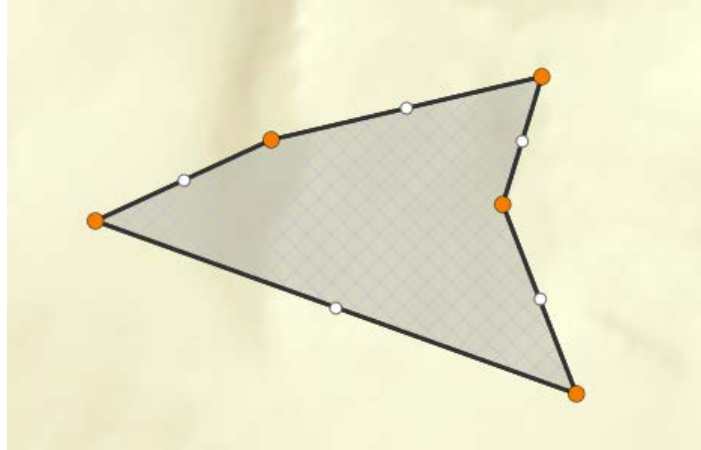


3. Use the **White** anchor points to edit existing line segments.
4. Press **Esc** on your keyboard when you are finished reshaping the anchor area.
5. Click **Yes** when you are asked to confirm that you want to reshape the anchor area.

Method 2



1. In the Anchor Area table below the map, hover over the anchor area that you want to change and click **Reshape**.
2. Use the **Orange** anchor points to move or resize connected line segments.



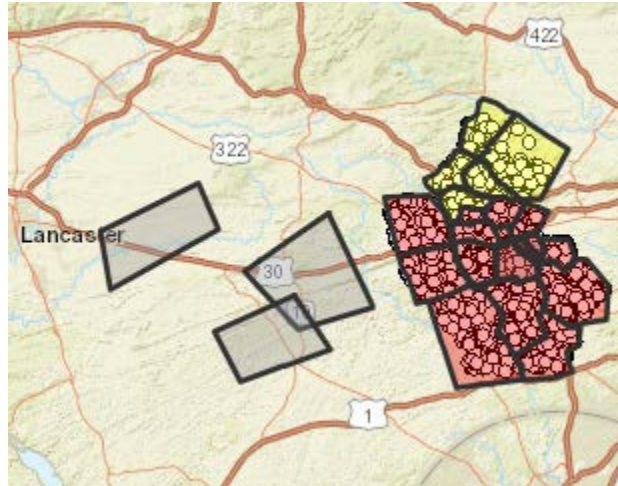
3. Use the **White** anchor points to edit existing line segments.
4. Press **Esc** on your keyboard when you are finished reshaping the anchor area.
5. Click **Yes** when you are asked to confirm that you want to reshape the anchor area.

Assigning Anchor Areas to Routes

1. Choose a route plan from the Route Plan dropdown.
2. In the Anchor Areas table below the map, click **Assign Anchor Areas**.

On the map you will see the anchor areas in your route plan. For each anchor area the polygon will be filled by red, yellow, or gray.

- **Yellow** anchor areas are assigned to the currently selected route.
- **Red** anchor areas are not assigned to the current route but are assigned to other routes.
- **Gray** anchor areas are not assigned to any route.



Note: The display may change as you choose different routes to view.

3. Choose the route you want to assign anchor areas to. You may work with only one route at a time.
4. Click a red or gray anchor area to assign it to the current route.
5. If you want to unassign an anchor area for the current route, click a yellow anchor area to unassign it. If it's not assigned to other routes it will turn gray; if it's also assigned to other routes, it will turn red.
6. When you are finished, click **Save**.
7. Click **Exit**.

Anchor Area Sequencing and Prioritizing

Anchor Area Prioritization enables you to choose which anchor areas should be serviced at specified places in a route (First, Last, or Sequenced). For example, if you have 10 anchor areas, and three are set to First, those three areas will be serviced first on the route, no matter where in the territory they are located.

Anchor Area Sequencing enables you to assign a specific sequence order to the anchor areas on your routes.

Priority Categories

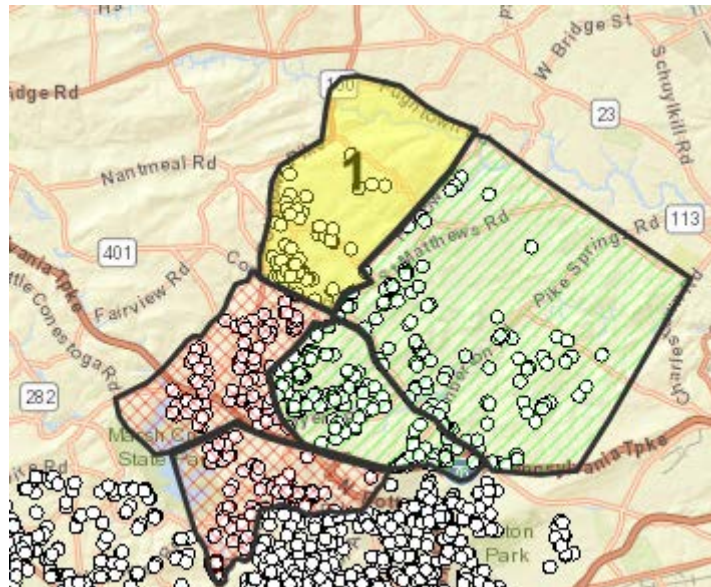
First - This category ensures that all stops in the specified area are serviced **before** all stops that are not in a First type anchor area. Multiple anchor areas may be designated as First, but there is no prioritizing anchor areas within a First category. The route may not complete all stops in one First anchor area before moving to other First stops, but

all First stops will be completed before moving to other categories. **First Stop Overrides** and **Time Window Stops** will take priority over First anchor area types.

Last - This category ensures that all stops in the specified area are serviced **after** all stops that are not in a Last type anchor area. Multiple anchor areas may be designated as Last, but there is no prioritizing anchor areas within a Last category. The route may not complete all stops in one Last anchor area before moving to other Last stops. **Last Stop Overrides** and **Time Window Stops** will take priority over Last anchor area types.

Sequence - This category ensures that anchor areas will be serviced according to their relative sequence value. All stops in a sequential type anchor area must be serviced before moving on to another anchor area. Stops that are not within a sequenced anchor area (in Unspecified areas or dynamically assigned) may be serviced at any point. You may assign up to 20 different sequence values per route. Multiple Sequence type anchor areas may have the same sequence value and will be serviced together, similar to how multiple First or Last areas are treated.

Unspecified - This category ensures that the specific area is included on a route, but it still allows the stops in that area to be sequenced as the solver determines is optimal. Stops in Unspecified anchor areas do not need to be sequenced as one grouping. Unless otherwise indicated, all anchor areas are considered Unspecified.



In the image above, the Eastern anchor areas (green fill) are designated as First type. The Northern anchor area is designated as Sequence with the sequence value displayed. The Western anchor areas (red fill) are designated as Last type.

Setting Priority and Sequence

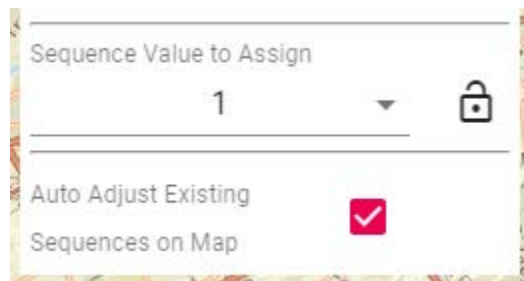
1. Click the Route Plan dropdown and choose the route plan you want to work with.
2. Click **Set Priority and Sequence** in the Anchor Areas table below the map.

3. Choose a route to view from the Routes list. The anchor areas assigned to the current route will be displayed on the map and will appear in the table below the map. If no anchor areas are shown, you will need to assign anchor areas to your route first.
4. You may set priority and sequence on the map or in the Anchor Areas table. Choose the method you prefer.

On the Map

You will use the buttons at the top of the table together with the anchor area polygons on the map to set Priority and/or Sequence.

1. To assign a **First** value to one or more anchor areas, click **Assign First** in the table.
2. Click one or more anchor areas on the map. The anchor area display will be updated to indicate it's a **First** type category.
3. To assign a **Last** value to one or more anchor areas, click **Assign Last** in the table.
4. Click one or more of your anchor areas on the map. The anchor area display will be updated to indicate it's a **Last** type category.
5. To assign a **Sequence** value to one or more of your anchor areas, click **Assign Sequence** in the table.
6. Click an anchor area on the map.
7. On the dialog at the right hand side of the screen, review the sequence information.



Sequence Value to Assign

1

Auto Adjust Existing Sequences on Map ☒

- **Sequence Value** - This section shows the next sequence value that will be used. If you want to assign a different sequence value to the next anchor area, click the dropdown and choose a different sequence number.
- **Lock Icon** - When the icon is unlocked (default) the sequence number will automatically advance. When the icon is locked, the sequence number will not advance, allowing you to assign the same sequence value to multiple anchor areas.
- **Auto Adjust Sequences** - When the setting is enabled, and a sequence value is assigned to an anchor area, the sequence value of all other anchor areas on that route will adjust up or down in value to make room for the new sequence value assignment. This prevents the user from being required to adjust the sequence value of all anchor

areas on a route when a new anchor area is introduced.

When the setting is disabled, and a sequence value is assigned to an anchor area, the sequence value of all other anchor areas on that route will NOT change. This is useful when the user wishes to assign the same sequence value to multiple anchor areas.

- 8. Continue adding priority and/or sequence assignments to your anchor areas as needed.
- 9. When you are finished, click **Save**.
- 10. Click **Exit**.

In the Table

You may edit the Priority and Sequence for your anchor areas using the Priority and Sequence fields in the table below the map.

Priority	Sequence
First ▼	▼
First ▼	▼
Last ▼	▼
Last ▼	▼
Sequential ▼	1 ▼

- 1. To set a priority for an anchor area, click the dropdown in the Priority field and choose the priority type to assign.
- 2. For Sequence type anchor areas, choose a sequence value to assign from the Sequence field.

*Note: If you have a First, Last, Unspecified, or blank value in the Priority field for an anchor area and choose a Sequence value, the Priority will be changed to **Sequenced** for that anchor area.*

- 3. Continue adding priority and/or sequence assignments to your anchor areas as needed.
- 4. When you are finished, click **Save**.
- 5. Click **Exit**.

Showing an Anchor Area on the Map

To show an anchor area on the map, hover over the anchor area you want to see and click **Show on Map**. The map will zoom to the chosen anchor area.

Editing an Anchor Area

The Edit Anchor Area tool enables you to edit all information about an anchor area at one time.

EDIT 

1. In the table below the map, hover over the anchor area you want to edit and click **Edit**.
2. On the Edit Anchor Area window that opens, you will see the anchor area name and label on the left side, and the anchor area assignment on the right side.
3. Edit the information about the anchor area as needed.
4. Use the **Add** button to add an anchor area assignment, for which you can assign route, priority, and/or sequence.
5. When you are finished, click **Save**. The table will be updated with the new information for the anchor area.

+

Deleting an Anchor Area

DELETE ANCHOR AREA

Are you sure you want to delete anchor area
Southwest ?

NO 

YES 

DELETE 

1. To delete an anchor area, hover over the anchor area in the list below the map and click **Delete**.
2. On the confirmation window, you will be asked to confirm that you want to delete the selected anchor area. This window will also inform you that it will be deleted from any route plans to which it is currently assigned. Click **Yes** to delete or **No** to return to the **Anchor Areas** page.

Using Dynamic Anchor Areas

Dynamic Anchor Areas allow a user to assign multiple routes of the same route type to a new or existing anchor area, and the stops within that anchor area will be distributed optimally between those selected routes. Users can also specify that stops be assigned to only one of those eligible work areas (Single) or be distributed between work areas (Distributed).

- **Single Work Area** – Stops within this Dynamic Anchor Area type will be assigned to only one of the eligible work areas, as deemed most optimal by the solver. Dynamic anchor area assigned the single option can be sequenced, but the sequencing options will only be utilized if the anchor area is assigned to that work area. For contingency stops, they will be assigned to only the work area that is selected, as any other non-dynamic anchor area would normally function.
- **Distributed** – Stops within this Dynamic Anchor Area type will be distributed between the eligible routes as optimally as possible. Anchor areas that are assigned the distributed type are unable to be sequenced and the option will be greyed out and/or unavailable throughout the application. Any contingency stops for the distributed dynamic anchor area type will be assigned to the closest work area based on proximity to an available stop, but only to a work area assigned as eligible in that area.

Enabling Dynamic Anchor Areas

Dynamic Anchor Areas are enabled at the route plan level, meaning you must enable them for each route plan in which you want to use them.

Note: Enabling Dynamic Anchor Areas will make interface changes to allow dynamic anchor areas to be created and used. It is recommended that you create a copy of your route plan before enabling dynamic anchor areas.

1. To enable Dynamic Anchor Areas for a route plan, click **Manage** in the menu and then click **Route Plans**.
2. On the Route Plans screen, hover over the route plan you want to use and click **Edit**.
3. On the route plan editing window that opens, click the **Use Dynamic Anchor Areas** checkbox.
4. A message will be displayed, warning you about the changes that will be made. Click **OK** to proceed.

Creating Dynamic Anchor Areas

There are multiple ways you can create a dynamic anchor area:

- Editing an anchor area (map or table)
- Assigning an anchor area
- Editing a route plan

Creating Dynamic Anchor Areas by Editing Anchor Areas

1. Click **Manage** in the menu and then click **Anchor Areas**.
2. On the Anchor Areas page, click the **Route Plans** dropdown and choose the route plan you want to use.

Note: The route plan must have Dynamic Anchor Areas enabled.

Option 1 – On the Map

1. Click one of the anchor areas on the map and click **Edit**.
2. On the **Edit Anchor Area** window that opens, click the Add button to add another work area of the same route type.
3. By default, all newly created Dynamic Anchor Areas will be designated as the **Single** type, but you can change it to the Distribute type by clicking the **Distribute Stops** checkbox.

Note: If you change the dynamic anchor area type to Distribute you will receive a message warning you that any sequences will be switched to Unspecified and that no sequencing is allowed for the Distribute type of dynamic anchor area.

4. When you are finished, click **Save**.

Option 2 – In the Table

1. In the Anchor Areas table, hover over one of your anchor areas and click **Edit**.
2. On the **Edit Anchor Area** window that opens, click the Add button to add another work area of the same route type.
3. By default, all newly created Dynamic Anchor Areas will be designated as the **Single** type, but you can change it to the Distribute type by clicking the **Distribute Stops** checkbox.

Note: If you change the dynamic anchor area type to Distribute you will receive a message warning you that any sequences will be switched to Unspecified and that no sequencing is allowed for the Distribute type of dynamic anchor area.

4. When you are finished, click **Save**.

Creating Dynamic Anchor Areas using Assign Anchor Areas

1. Click **Manage** in the menu and then click **Anchor Areas**.
2. On the Anchor Areas page, click the **Route Plans** dropdown and choose the route plan you want to use.

Note: The route plan must have Dynamic Anchor Areas enabled.

3. Click **Assign Anchor Areas**. You will see anchor areas assigned to the current route indicated in yellow, unassigned in gray, and assigned to other routes in red.
4. Choose a route to work with.
5. Click one of the gray or red anchor areas. Continue clicking to cycle through the various assignments. To create a Dynamic Anchor Area, choose the **S** or **D** type assignment.

Note: Refer to the legend for a reminder of what the various assignment indicators mean.

6. Continue for the other routes in your route plan as needed.
7. When you are finished, click **Save** and then click **Exit**.

Creating Dynamic Anchor Areas by Editing a Route Plan

1. Click **Manage** in the menu and then click **Route Plans**.
2. On the Route Plans page, hover over the route plan you want to use and click **Edit**.

Note: The route plan must have Dynamic Anchor Areas enabled.

3. On the **Edit Route Plan** window that opens, hover over the work area you want to edit and click **Edit**.
4. On the **Assign Anchor Areas** window, you will see that the available anchor areas are categorized by **Unassigned** (not assigned to any routes) and **Assigned** (Assigned to other routes than the current one).
5. To create a Dynamic Anchor Area, drag one or more of the Assigned anchor areas to the categories on the right side of the window. Any existing or created Dynamic Anchor Areas are outlined in green.

Note: By default, the type will be Single but you may click the text to toggle to the Distribute type.

6. When you are finished, click **Save**.

Sequencing Dynamic Anchor Areas

Dynamic Anchor Areas marked as **Single** type may be sequenced. Sequencing options will be grayed out for Distribute type anchor areas.

Note: If you sequence a Single type and then change it to a Distribute type, the sequences will be lost.

There are three ways to sequence Dynamic Anchor Areas:

- Editing an anchor area
- Editing a route plan
- Using Set Priority and Sequence

Sequencing Dynamic Anchor Areas on the Anchor Area Edit Window

1. Click **Manage** in the menu and then click **Anchor Areas**.
2. On the Anchor Areas page, click the **Route Plans** dropdown and choose the route plan you want to use.

Note: The route plan must have Dynamic Anchor Areas enabled.

Option 1 – On the Map

1. Click one of the anchor areas on the map and click **Edit**.
2. On the **Edit Anchor Area** window that opens, use the dropdowns under **Priority** to set the priority group or sequence value.
3. When you are finished, click **Save**.

Option 2 – In the Table

1. In the Anchor Areas table, hover over one of your anchor areas and click **Edit**.
2. On the **Edit Anchor Area** window that opens, use the dropdowns under **Priority** to set the priority group or sequence value.
3. When you are finished, click **Save**.

Sequencing Dynamic Anchor Areas on the Route Plan Edit Window

1. Click **Manage** in the menu and then click **Route Plans**.
2. On the Route Plans page, hover over the route plan you want to use and click **Edit**.

Note: The route plan must have Dynamic Anchor Areas enabled.

3. On the **Edit Route Plan** window that opens, hover over the work area you want to edit and click **Edit**.
4. As you move anchor areas to the right side, you may place them in priority groups or in sequence order.
5. When you are finished, click **Save**.

Sequencing Dynamic Anchor Areas using Set Priority and Sequence

1. Click **Manage** in the menu and then click **Anchor Areas**.
2. On the Anchor Areas page, click the **Route Plans** dropdown and choose the route plan you want to use.

Note: The route plan must have Dynamic Anchor Areas enabled.

3. Click **Set Priority and Sequence** in the Anchor Areas table below the map.

You may set priority and sequence on the map or in the Anchor Areas table. Choose the method you prefer.

On the Map

You will use the buttons at the top of the table together with the anchor area polygons on the map to set Priority and/or Sequence.

1. To assign a priority group to an anchor area, click the button in the table for the type you want to assign (First or Last) and then click an anchor area on the map.
2. To assign a Sequence value to one or more of your anchor areas, click **Assign Sequence** in the table.
3. Click an anchor area on the map.
4. On the dialog at the right hand side of the screen, review the sequence information. You may change the sequence number if needed.
5. Continue adding priority and/or sequence assignments to your anchor areas as needed.
6. When you are finished, click **Save**.
7. Click **Exit**.

In the Table

You may edit the Priority and Sequence for your anchor areas using the Priority and Sequence fields in the table below the map.

1. To set a priority for an anchor area, click the dropdown in the **Priority** field and choose the priority type to assign.
2. For **Sequence** type anchor areas, choose a sequence value to assign from the **Sequence** field.

Note: If you have a First, Last, Unspecified, or blank value in the Priority field for an anchor area and choose a Sequence value, the Priority will be changed to Sequenced for that anchor area.

3. Continue adding priority and/or sequence assignments to your anchor areas as needed.
4. When you are finished, click **Save**.
5. Click **Exit**.

Managing Stop Overrides

For any stop, you may set specific requirements that are different from most of the other stops on your route. These requirements include:

- Whether a stop must be placed on a certain route or route type
- Whether a stop needs to be serviced first or last on a route
- The preferred load section for the stop

These preferences can be applied to both pickup and delivery stops, or just pickup or just delivery. Stop overrides in DRO let you assign these requirements to stops in your work areas.

When you are creating a stop override, you will decide the **Route Designation**, **Stop Order**, and the **Load Section or Pickup Time Window**. The options for each are below:

Route Designation

Route Designation

☒ N/A

☐ Route Type ▼

☐ Route Name ▼

- **No Route Designation** – Only allowed if a Stop Order assignment is made
- **Route Type** – The stop may be assigned to a Bulk or LP route, meaning it can be placed on any route of the chosen category
- **Route Name** – The stop will be placed on the work area you specify

Stop Order

Stop Order

☒ No Preference

☐ First Stop

☐ Last Stop

- **No Preference** – The stop will be placed in the most efficient order on the route.
- **First** – The stop will be serviced as the first stop on the route
- **Last** – The stop will be serviced as the last stop on the route

*Note: If there are multiple stops on a route marked as **First Stop** or **Last Stop**, those stops will be done in the most efficient order, still at the beginning or end of the route as required.*

Load Section

Load Section

☒ No Preference

☐ Shelf ▼

☐ Floor ▼

- **No Preference** - The items for that stop will be placed in the vehicle according to the sequence order of the stop.
- **Shelf** - The items for the stop will be placed on the shelf section you choose.
- **Floor** - The items for the stop will be placed in the floor section you choose.

Pickup Time Window

☒ Pickup Time Window ?

Pickup Opening At Or After ▼

And Closing Before Or At ▼

This optional feature enables a pickup to be assigned to a route based on its specific time window, so that multiple pickups at one location may receive different route assignments. A pickup window must be fully encompassed within the time window specified in the stop override to be respected. This option is available when Stop Type is set to Pickup Only.

Adding a Stop Override

ADD +

1. To add a stop override, click **Add** on the **Stop Override** page.
2. On the **Candidate Address Records** window that opens, enter an address and/or ZIP code to find addresses and click **Search**. The list of stops with that address or in that ZIP code will be displayed.

Candidate Address Records

Candidate Address

Search

	Stop Name	Stop Address	Postal Code
☐	KARL ENVIRONMENTAL GROUP	20 LAUCK RD	
☐	JAMES HARRISON	600 IMPERIAL DR	
☐	DAVID HINKS	48 E SUMMIT ST	
☐	HENRY GIROUX	33 OREGON RD	
☐		101 LISA CT	
☐	MARC PETERS	1 SYLVAN LN	
☒	BERGS CANVAS GALLERY	477 ALLEGHENYVILLE RD	
☐	ASHLEY ENGELMAN	290 BLIMLINE RD	
☐	DEB RISSMILLER	295 LEDGE ROCK RD	

CANCEL

CONFIRM

3. Scroll down the list or use the search field to enter search criteria to narrow down the list of candidate addresses. The list will update as you enter your search criteria.

*Note: You may also use the selection tool on the map to select one or more stops on the map to populate the **Candidate Addresses** list.*

4. When you see the address you want to use, click the radio button next to the stop name and click **Confirm**. The **Add Stop Override** window will open and you will see the Name, Address and ZIP information for the stop.

ADD STOP OVERRIDE

Apply this override to Stop Type:

☒ Both Pickup and Delivery
 ☐ Pickup Only
 ☐ Delivery Only

Route Designation

☒ N/A
 ☐ Route Type
 ☐ Route Name

Stop Order

☒ No Preference
 ☐ First Stop
 ☐ Last Stop

Load Section

☒ No Preference
 ☐ Shelf
 ☐ Floor

CANCEL

SAVE

5. In the **Apply this override to Stop Type** section, choose whether the override applies to both **Pickup and Delivery** stops, **Pickup** stops only, or **Delivery** stops only.

Note: Stop type overrides need to match the stop type of the data presented in DRO. To split a combination stop into separate pickup and delivery stops, contact your station to set up a delivery time window.

6. In the **Route Designation** area, choose N/A, assign the stop to a type of route (Bulk or LP), or assign the stop to a specific route. If designating the stop override to a specific type, please note the following: If **Bulk** type is selected, the stop will be assigned to any bulk route designated in your active route plan. If **LP** route type is selected, the stop will be assigned to any LP route designated in your route plan.
7. In the **Stop Order** area, choose No Preference, First Stop, or Last Stop.
8. If you have chosen **Delivery Only** or **Pickup and Delivery** for Stop Type, in the **Load Section** area, choose No Preference, assign the stop to a shelf section, or assign the stop to a floor section. This is not applicable for Pickup overrides.
9. For **Pickup** stops only (Optional): The Pickup Time Window feature enables a pickup to be assigned to a route based on its specific time window, so that multiple stops at one location may receive different route assignments. A pickup window must be fully encompassed within the time window specified in the stop override to be respected. Choose the start and end times from the dropdowns to set up a time window for this stop.
10. When you are finished, click **Save**.
11. To activate the stop override that was just created, select the route plan(s) it should be enabled in on the Enable Stop Override window.

Note: This step can also be completed within the route plan itself.

12. Click **Save**.

The stop name will be displayed in the **Stop Override Master Table** below the map.

When you are finished adding your stop overrides, use the **Edit Route Plan** tool on the **Route Plans** management page to add them to the route plan(s) for which they should be used, if you did not assign them after adding them.

Adding a Stop Override on the Map Page

1. Click **Map** in the side menu to go to the Map page.
2. On the Map page, choose the Dispatch and Solution you want to use.
3. If you know which route the stop you want to use is on, you can choose to view only that route. The Stop Information table will show stops for only the route you chose.
4. Use the checkbox(es) in the table to select the stop(s) you would like to create an override for. If multiple stops are selected, a unique override will be created for each in the subsequent steps.
5. Click the **Create Stop Override** button.
6. Choose the type of stop to which you want the override assigned.



Note: Stop type overrides need to match the stop type of the data presented in DRO. To split a combination stop into separate pickup and delivery stops, contact your station to set up a delivery time window.

7. Choose either a Route Type or Route Name, or leave N/A. If designating the stop override to a specific route type, please note the following: If **Bulk** type is selected, the stop will be assigned to any bulk route designated in your active route plan. If **LP** route type is selected, the stop will be assigned to any LP route designated in your route plan.
8. Choose a Stop Order preference or leave as **No Preference**.
9. Choose either a Shelf or Floor section or leave as **No Preference**.
10. For **Pickup** stops only (Optional): The Pickup Time Window feature enables a pickup to be assigned to a route based on its specific time window, so that multiple stops at one location may receive different route assignments. A pickup window must be fully encompassed within the time window specified in the stop override to be respected. Choose the start and end times from the dropdowns to set up a time window for this stop.
11. When you are finished, click **Save**.
12. To activate the stop override that was just created, select the route plan(s) it should be enabled in on the Enable Stop Override window.

Note: This step can also be completed within the route plan itself.

13. Click **Save**.

Editing a Stop Override

EDIT STOP OVERRIDE

Stop Name

Stop Address

Postal Code

16314

Apply this override to Stop Type:

☒ Both Pickup and Delivery

☐ Pickup Only

☐ Delivery Only

Route Designation

☐ N/A

☐ Route Type

☒ Route Name

Stop Order

☐ No Preference

☒ First Stop

☐ Last Stop

Load Section

☒ No Preference

☐ Shelf

☐ Floor

CANCEL

SAVE

1. To edit a stop override, hover over the override you want to change on the Stop Overrides page and click **Edit**.

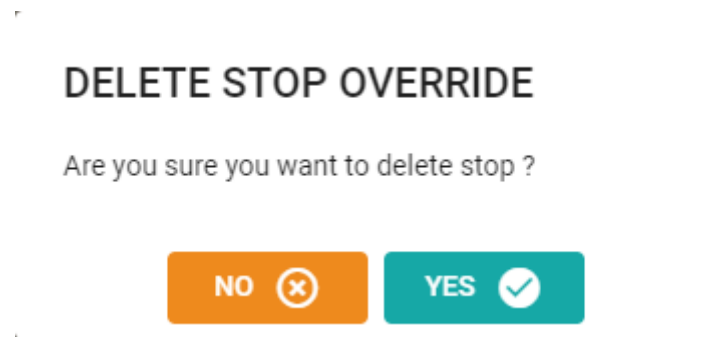
EDIT

2. On the **Edit Stop Override** window that opens, make the changes you want to the stop override.
3. When you are finished, click **Save**.
4. To activate the stop override that was just edited, select the route plan(s) it should be enabled in on the Enable Stop Override window.

Note: This step can also be completed within the route plan itself.

5. Click **Save**.

Deleting a Stop Override



1. To delete a stop override, hover over the override you want to delete and click **Delete**.
2. The confirmation window will tell you the name of the stop override you are deleting and whether it is assigned to any route plans. Click **Yes** to delete, or **No** to return to the main **Stop Override** page.

Exporting Stop Overrides to csv

If needed, you may export the list of stop overrides to a csv file.

1. Click **Manage** and go to the **Stop Overrides** page.
2. In the table below the map, click the Export to CSV button.

The stop overrides table will be exported and downloaded according to your browser settings.

Managing Street Settings

The Street Settings section enables you to control how streets in your CSA are used.

Street Settings Tools



Zoom In – Zooms the map in each time you click it



Zoom Out – Zooms the map out each time you click it



Default Map View – Returns the map to the default view



Show/Hide Delivery Side Layer - Shows or hides the delivery side indicators on the streets.



Show/Hide Closures Layer - Shows or hides the street closures indicators on the streets.



Show/Hide Street Layer - Shows or hides the streets only

Delivery Side

The Delivery Side tool lets you view the delivery side (either right side or both sides) of the streets in your CSA. You may easily change the delivery side of streets as needed.

Delivery Side Legend

Legend

Delivery Side

Right Side Only

Both Sides

Legend - Shows the colors of the delivery side indicators and what each means

Viewing and Changing Delivery Side

1. To view and/or change the delivery side designations of your streets, click **Street Settings** in the **Manage** menu on the left side of the screen.

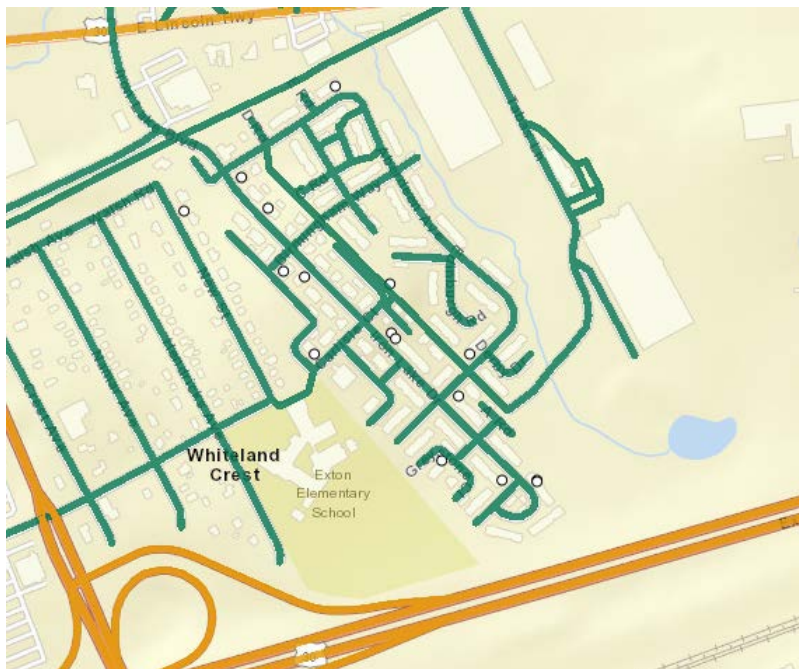


2. When you first see the screen, your map will be zoomed out to show your whole service area. Click the **Show delivery side layer** button.

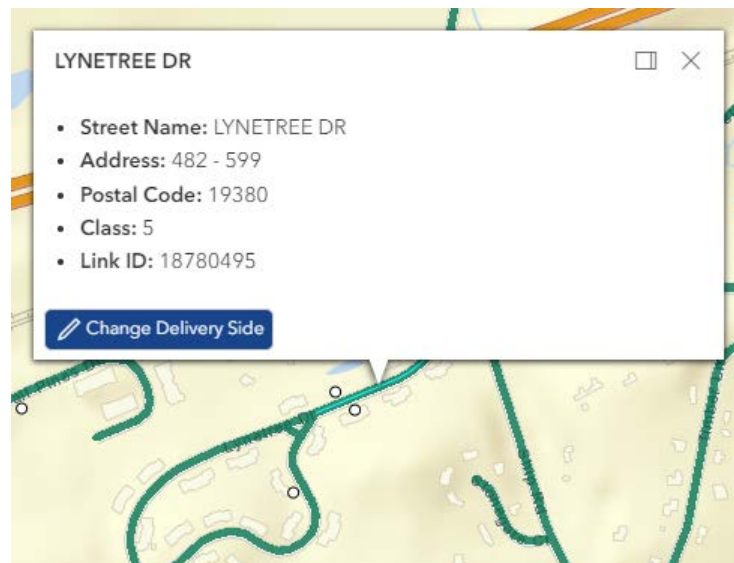
3. You will see a message telling you to zoom in to activate street editing. Click **OK** on the message to continue.



4. Use the **Zoom In** button (or your mouse scroll wheel) to zoom in closely to the streets. When you are close enough, the orange and green indicators will appear. The legend will also be populated with what each color means.



5. Click one of your streets. A window will appear, displaying information about the street segment you clicked.



6. If you want to change the delivery side of the segment you are viewing, click the **Change Delivery Side** button on the information window. The street will change to the other type of delivery side.



Note: There are some types of streets—like major highways—that cannot be changed. You will see a note on the information window letting you know whether the street you are viewing is one of these streets.

The changes you make will be used the next time you generate routes.

Street Closures

Street closures can be used to discourage travel on a street. When a street is closed, DRO will avoid using it unless traveling on the street is required for making a delivery or pickup.

Street Closures Legend

Legend

Closures



Legend - Shows the colors of the street closure indicators and what each means

Managing Street Closures

1. To view and/or change the closures on your streets, click **Street Settings** in the **Manage** menu on the left side of the screen.

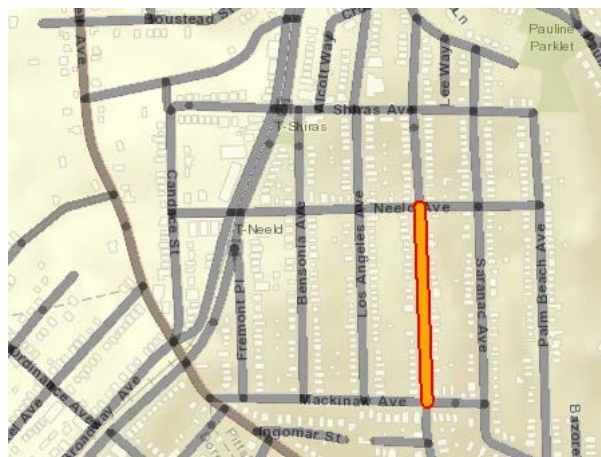


2. When you first see the screen, your map will be zoomed out to show your whole CSA. Click the **Show closures layer** button.

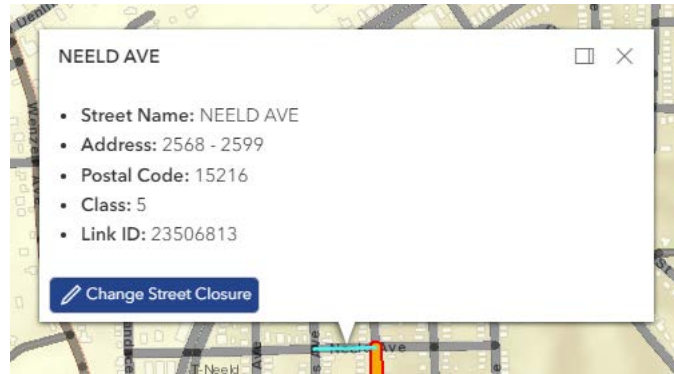
3. You will see a message telling you to zoom in to activate street editing. Click **OK** on the message to continue.



4. Use the **Zoom In** button (or your mouse scroll wheel) to zoom in closely to the streets. When you are close enough, the street closure indicators will appear. The legend will also be populated with what the color means.



5. Click one of your streets. A window will appear, displaying information about the street segment you clicked.



6. If you want to change the closure of the segment you are viewing, click the **Change Street Closure** button on the information window. The street will change to Closed.



7. If you want to open a closed street, click the **Change Street Closure** button again to open it.

The changes you make will be used the next time you generate routes.

Viewing Streets Only

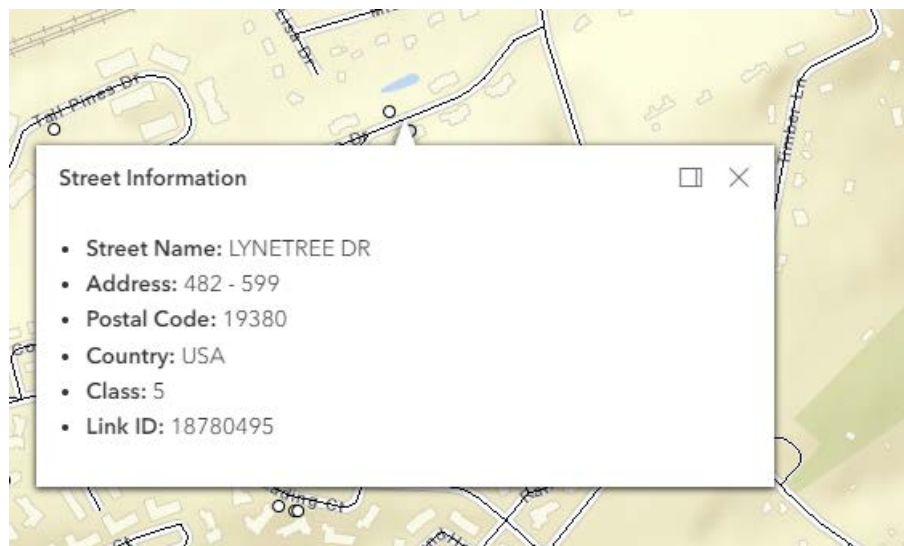
If you want, you may view just the streets without the delivery side or closure indicators.



1. To view streets only, click the **Show Street Layer** button. If you were viewing the delivery side or closure layer, it will be removed when you click the Show Street Layer button.



2. Click one of the streets in your area. An information window will appear, showing details about the street segment you clicked.



Note: You cannot change delivery side or closures while viewing only the street layer.

Managing Defaults

Editing Bulk Stop Customization

Route Type Rules: Bulk Routes

STOP CONTAINS: More than 10 packages
and/or 15 cubic feet

CANCEL 

SAVE 

EDIT 

1. To edit Bulk stop customization, hover over the Bulk stop definition and click **Edit**.
2. Enter the package threshold and/or the cubic foot threshold for a stop to be considered a bulk stop. If either of the values on this screen is met, a stop will be considered a bulk stop and placed on a bulk route type. It is not necessary that both are met.

Note: If Bulk Routes are not being utilized, the default values can be kept here.

3. Click **Save**. The updated route type definition will be used the next time routes are generated.

Editing Large Package Customization

The characteristics of Large Packages can be customized in DRO for the purposes of package sortation to service provider-designated LP work areas. Keep in mind that the LP characteristics applied in DRO for sortation and preload have no bearing on the Large Package definition that is listed on the service provider's MyGroundBizAccount. Only the LP definition on MGBA is used to determine the service provider's Large Package Mix Percentage for the purposes of the Large Package Mix Charge. Settings and LP metrics in DRO have no impact on the LP definition or metrics in MGBA.

LARGE PACKAGE CUSTOMIZATION

	☐ Use Custom Defaults	☒ Use FedEx Defaults
Weight is greater than	50 pounds	50 pounds
OR Single dimension greater than	48 inches	48 inches
OR Length (longest side) + Girth [(width x 2) + (height x 2)] greater than	105 inches	105 inches
☒ Service Nearby Regular Stops		
☒ Service Excess Bulk Stops		
CANCELSAVE		

EDIT 

1. To edit large package customization, click **Edit** on the right side of the Large Package section.
2. On the Large Package Customization window that opens, choose whether you are using FedEx Defaults or Custom Defaults. If you are using FedEx Defaults, skip to step 4.
3. If you are using custom defaults, click the Custom Defaults checkbox and enter the dimensions that define a Large package. A stop will be considered an LP stop if at least **one package** meets **one or more** of the criteria you use. All three **do not** have to be met to be considered a Large Package.
4. Choose whether LP routes can service nearby regular stops. Nearby means the regular stop is located on the same street segment (the same block).
5. Choose whether LP routes are allowed to service excess bulk stops. If there are more bulk stops than bulk route capacity, indicate if the LP routes can service the excess bulk.
6. Click **Save**. The updated LP definition will be used the next time routes are generated.

Editing Small Stop Customization

SMALL STOP CUSTOMIZATION

	DRO Defaults	Current Settings
Number of packages at stop is less than or equal to	2	2 packages
AND total cube of stop is less than	1	1 cubic feet
AND weight of every package at stop is less than	10	10 pounds
AND every single package side dimension at stop is less than	15	15 inches
Exclude Hazmat Stops from Small Routes	☐	
Exclude CDO Stops from Small Routes	☐	

RESET DEFAULTS

CANCEL

SAVE

EDIT 

1. To edit Small stop customization, click **Edit** on the right side of the Small Package section.
2. On the Small Stop Customization window that opens, enter the values that define a Small stop. The DRO defaults are in the first column.
3. Use the checkboxes if you want to exclude Hazmat and/or CDO stops from small routes.
4. If you need to, you can reset default using the button in the bottom left of the window.
5. When you are finished, click **Save**.

Editing Floor Loading Rules

EDIT FLOOR LOADING RULES

STOP CONTAINS:

One or more package(s) with a weight greater than	50	pounds
AND/OR a length greater than	32	inches
AND/OR a cube greater than	15	cubic feet

CANCEL 

SAVE 



1. To edit floor loading rules, click **Edit** in the **Floor Loading Rules** section on the **Defaults** page.
2. On the **Edit Floor Loading Rules** window that opens, make changes to the floor loading rules you want to edit.
3. When you are finished, click **Save**.

Adding Driver Breaks

Add/Edit Breaks

Break #

3

Earliest Start Time

11:00

Latest Start Time

12:30

Length of Break

00:30

☐ Enabled

CANCEL 

SAVE 



1. To add a driver break, click **Add** in the Driver Breaks section on the Defaults page.
2. On the **Add/Edit Breaks** window that opens, enter a **Break #**, **Earliest** and **Latest Start Times**, and **Length of Break**.

Note: Earliest start time, latest start time, and length of break must be in HH:MM format.

3. Click the **Enabled** checkbox if this break is currently an active break.
4. When you are finished, click **Save**. Your break will be added to the list of breaks on the main page.

Editing Driver Breaks

Add/Edit Breaks

Break #
3

Earliest Start Time
11:00

Latest Start Time
12:30

Length of Break
00:30

☐ Enabled

CANCEL 

SAVE 

EDIT 

1. To edit a driver break, hover over the break you want to change and click **Edit**.
2. On the **Add/Edit Breaks** window, make changes to the break information as needed.
3. When you are finished, click **Save**.

Deleting a Driver Break

DELETE 

1. To delete a driver break, hover over the break you want to delete and click **Delete**.
2. You will be asked to confirm whether you want to delete the break. Click **Yes** to delete, or **No** to return to the main page.

Enabling Relays

Note: If you do not see the Relays section, relays are not enabled for your station.

1. On the **Defaults** page, scroll down to the **Relays** section.
2. Click the checkbox to enable relays.

Viewing Historical Volume Information

The Historical Volume page provides historical volume information to assist in providing context for volume on a similar day of week or a custom previous date. This section will allow a user to select any sort date from the previous 13 months and view work area level detail, as well as summary information compared to current data visible. This feature can be beneficial for ad-hoc user reference, as well as work area level planning.

1. To view Historical Volume information, click Historical on the menu on the left side of the screen.
2. Click the Date Selection calendar icon to choose a date to view. You may also enter the date manually on the Date Selection line.

The Day Summary and Work Area Summary tables on the Historical Volume page will be updated, including a small table in the upper right of the screen showing a comparison between the selected date and the current date. The Day Summary and Work Area Summary tables may be exported to csv files using the Export to CSV buttons for each.

Using the Sandbox Environment

The Sandbox environment in DRO enables you to model routing scenarios on actual data without affecting your daily routing.

Accessing the Sandbox Environment

1. To access the Sandbox environment, click Switch View at the top right of your screen.
2. You will see three options: Sandbox View, Sort Plan View, and Normal View. Click **Sandbox View**.
3. You will see a warning about entering Sandbox view. Click **OK** to proceed.

A configuration settings window will open. On this window, you can confirm and customize the data visible in the Sandbox.

The options available on the configuration screen include:

- **Select day to model** – Enables you to use the current day's data, or import stop data from the most recent previous day of week to model day-specific activity
- **Dispatch mode** – If you are operating in a station configured to run multiple dispatches, this option allows you to view and manage data separately (Dual) or in a combined view (Single)
 - If Sandbox geography has been added specific to a dispatch, that will be displayed on the 'Dispatch' column of the table
 - If a Dispatch has not been specified for a Sandbox geography rule, the 'Dispatch' column will display 'Default'
- **Custom Sandbox Geography** – If custom geography has been added and enabled to your sandbox, such as the inclusion of additional postal codes, geographic polygons, or CSAs, you may now choose to include that data by checking the **Enabled** option, or exclude that data by deselecting the **Enabled** checkbox
- **Sandbox Rules Summary** – A summary of the rules being applied to your sandbox will be displayed in the **Sandbox Rules** summary at the top of the Sandbox mode dashboard
 - To modify those rules after they have been confirmed, click the Sandbox Rules **Edit** button on the Sandbox Dashboard page to return to the configuration screen

While you are in Sandbox view, every page will display a yellow banner at the top to differentiate Sandbox view from your normal view. When you are finished, click **Normal View** at the top right of your screen.

Syncing from Production

The **Sync from Prod** feature enables you to import existing data so that the Sandbox mirrors the current live environment.

Note: When Sync from Prod is used, the current data in Sandbox that is not copied to the live environment will be overwritten.

1. Click Sync from Prod on the menu on the left side of the screen.
2. A summary window will open, showing details about the data in the live environment and in the Sandbox environment. Syncing from production will replace existing Sandbox data. Click **Confirm** to sync your data.

Copying a Route Plan

A route plan is a template that can be created and saved for future use. It contains the unique combination of route parameters to be used on a given day (i.e., vehicles, anchor areas, stop overrides).

For unique parameters created in the Sandbox to be used in the Live Environment, they must be attached to a route plan.

Route plan names must be unique to ensure active DRO plans are not overwritten. If Route Plan name is not unique you will be prompted to rename the Route Plan.

1. Click **Copy Route Plan** on the menu on the left side of the screen.
2. On the Copy Route Plan page, click the dropdown and choose a Route Plan that you want to copy.
3. Click **Confirm**.
4. A summary window will display details about the route plan you are copying. Click **Copy**.

Adding New Work Areas

1. To add a new work area, click **Manage** on the menu on the left side of the screen and click **Fleet**.



2. On the Vehicle Management tab, click **Add**.
3. On the **Add Vehicle** window, click the **Work Area Name** dropdown and choose a work area. The Work Area Names are populated from the Work Area Master Table, along with the current Work Area Number. For the Sandbox environment, once all available Work Areas are utilized, the user can type in the name of an additional Work Area. Use the drop-down to select an available Work Area Name.

Note: All work areas, including additional ones created, must be an available work area entered into the Work Area Master Table or you will be unable to copy the route plan from the Sandbox to live DRO. Once a vehicle is assigned to a work area not in the Work Area Master Table, the work area will be highlighted in red in the Vehicle Management table.

*Note: **Work Area** is required.*

4. Enter a **Vehicle ID**. Vehicle ID can be alpha and/or numeric.
5. Click the **Vehicle Type** dropdown and choose the type of vehicle. If the Vehicle Type is not listed, it will need to be added in the Fleet section under Vehicle Cost Profile.
6. **Vehicle Capacity** is automatically populated when you choose the **Vehicle Type**, but you may change it if needed. This should be the actual, demonstrated capacity within the vehicle, which by default is 60% of actual vehicle cube. If not otherwise dictated through Anchor Areas or Stop Overrides, a vehicle will not be loaded beyond this stated capacity.
7. Click the **Route Type** dropdown and choose the type of route this vehicle services.
 - **Bulk Routes** - Bulk route rules can be defined in the Defaults tab. A stop will be considered a bulk stop and assigned to this route type if it exceeds either the package minimum or cubic foot minimum that is defined under **Bulk Stop Customization** in the Defaults tab of DRO.
 - **LP Routes** - Large Package routes will only be eligible to service packages specifically as LP. Additional preferences for LP routes can be configured under **Large Package Customization** in Defaults.
 - **REG Routes** - Regular routes do not consider user-defined route type rules.
 - **SM Routes** - Small routes can be defined in the Defaults tab. a stop will be considered a Small stop if it meets the criteria defined under **Small Stop Customization**.
8. **Front to Back Assignment** indicates whether the cargo area can be accessed through the front of the vehicle. If selected, packages will be loaded with the first stops in sequence toward the front of the vehicle and last stops toward the back, unless otherwise dictated through a Stop Override. If not selected, packages will be loaded with the last stops in sequence toward the front of the vehicle and the first stops toward the back, unless otherwise dictated through a Stop Override.
9. Click the **Load Profile** dropdown and choose a load profile. The load profile can be modified here from the default selection of **Section** loading. Please note that when **No Section** is selected, packages will not receive shelf assignments and will indicate to the loader to stage packages, as opposed to van loading.

Note: You will see a description of the Load Profile you chose below the dropdown.

10. If you choose Section as your load profile, the **Section Avail** button will become available. Section availability enables you to customize the section configuration for the vehicle. See **Customizing Section Availability** above for more information.
11. Click **Save** when you are finished.

Using Sort Plan View

Accessing Sort Plan View

1. To access Sort Plan View, click Switch View while you are in the regular DRO environment.
2. On the view options page, click Sort Plan View.
3. A warning will appear about what you will see in the Sort Plan view. Click OK. to proceed.

While you are in Sort Plan view, every page will display a yellow banner at the top to differentiate Sort Plan view from your normal view. When you are finished, click **Normal View** at the top right of your screen.

Using the Manifest View

DRO Manifest View is a viewing mode that will display the state of routes as they exist at the time of the final manifest request from the STAR application ("Final Manifest Request" refers to an actual manifest generated from the STAR application and not the Manifest Preview application for preliminary manifests).

In this view, routes that have generated a manifest for that sort date will display along with the time stamp of that request. Routes that have not requested a manifest, or are unchanged from the original sort plan, will display the state of the route at the time of sort plan submission, and will not display a time stamp.

This view can be used to:

- View the final state of routes throughout the morning dispatch
- Compare any changes to routes from when the sort plan was submitted
- View specific stops that are inserted into a route
- View the type of manifest that was requested (insert vs. resequence)
- View the approximate status of vehicle dispatches

Accessing Manifest View

1. To access the Manifest View, click Switch View while you are in the regular DRO environment.
2. On the view options page, click Manifest View.
3. A warning will appear about what you will see in the Manifest view. Click OK to proceed.

While you are in Manifest view, every page will display a yellow banner at the top to differentiate Manifest view from your normal view. When you are finished, click **Normal View** at the top right of your screen.

Exploring the Map in Manifest View

1. Click **Map** in the menu on the left side of the screen.
2. On the Map page, click the **Manifest** dropdown and choose the view you want to see:
 - **Sort Plan + Manifest Jobs** –Will display all routes, regardless of manifest status
 - **All Manifests**–Will only display routes which have requested a manifest for that sort date
 - **Normal Manifests**–Will only display routes that have generated a manifest using the standard 'insert' option

- **Resequenced Manifests**–Will only display routes that have generated a manifest using the 'resequence' option
- **Custom Selection**–Will display in the dropdown if the user has manually selected a subset of routes to display

Routes will display based on most recent manifest request, along with the time stamp of that request. Routes without time stamps have not yet requested a manifest (or have no changes from the original sort plan) and will display the route configuration from the time of the sort plan submit.

3. In the table below the map, you may click the **Show All** dropdown to choose to filter the stops to show **Insert** or **Pickup** stops.

*Note: The **Insert** option will highlight any stops in the manifest request that have updated since the sort plan submission. This could include stops that were not present in the sort plan, where information has been updated, or those that have been dock flexed to a different work area prior to requesting a manifest*

Exploring the Reports Page in Manifest View

The Reports page in Manifest View is similar to the Reports page in Normal View, with fewer reports displayed. The Stop Detail report, Cost Detail report, and Time and Distance Detail report are available in Manifest View. For details about these reports, see the Reports section earlier in the DRO User Guide.